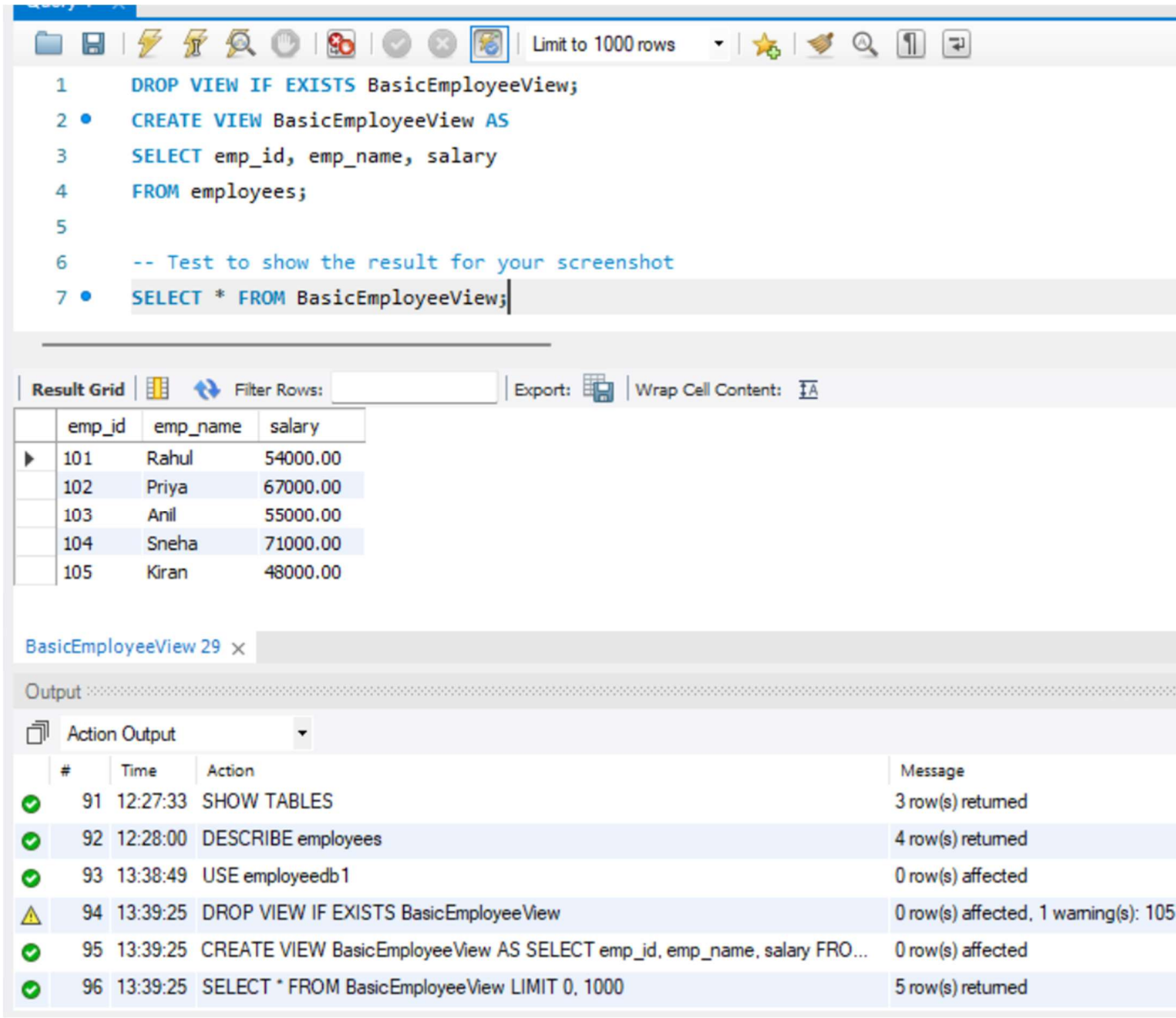
MYSQL VIEWS

Practical Examples and Student Exercises using MySQL Workbench

1.What is a View?

A **View** is a **virtual table** created from one or more existing tables.

- It **does not store data** itself.
- It stores only the **SQL query**.
- Whenever you query the view, MySQL retrieves the latest data from the underlying tables.



2. Setup: Employees and Departments Tables

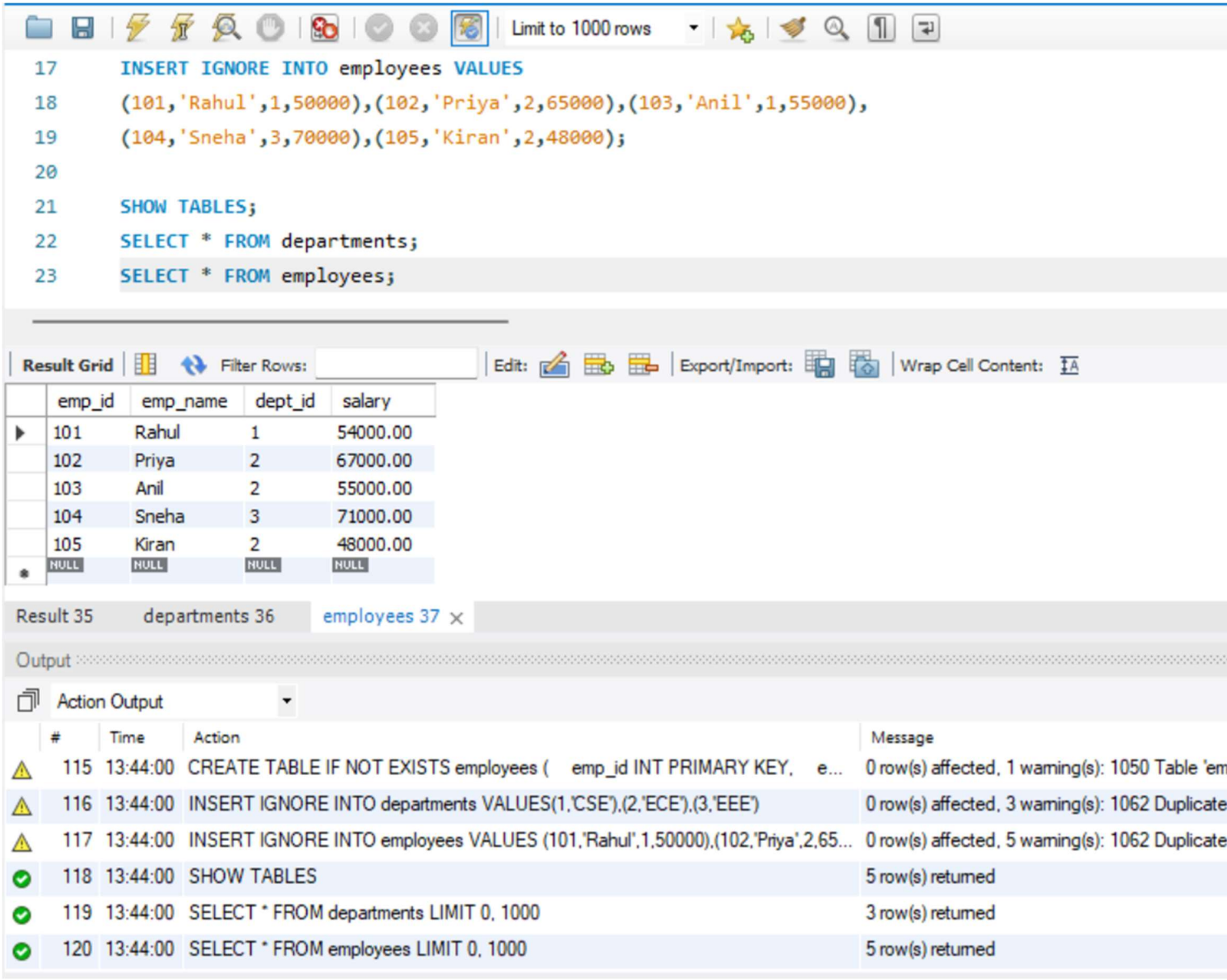
Run this setup first. If these tables already exist with data, you may skip the setup.

```
CREATE DATABASE IF NOT EXISTS employeedb1;
USE employeedb1;

CREATE TABLE IF NOT EXISTS departments (
   dept_id INT PRIMARY KEY,
   dept_name VARCHAR(50)
);

CREATE TABLE IF NOT EXISTS employees (
   emp_id INT PRIMARY KEY,
   emp_name VARCHAR(50),
   dept_id INT,
   salary DECIMAL(10,2)
);

INSERT IGNORE INTO departments VALUES
(1,'CSE'),
(2,'ECE'),
(3,'EEE');

INSERT IGNORE INTO employees VALUES
(101,'Rahul',1,50000),
(102,'Priya',2,65000),
(103,'Anil',1,55000),
(104,'Sneha',3,70000),
(105,'Kiran',2,48000);
```

3. Verify the Tables

```
SHOW TABLES;

SELECT * FROM departments;
```

SELECT * FROM employees;



```
17  INSERT IGNORE INTO employees VALUES
18  (101,'Rahul',1,50000),(102,'Priya',2,65000),(103,'Anil',1,55000),
19  (104,'Sneha',3,70000),(105,'Kiran',2,48000);
20
21  SHOW TABLES;
22  SELECT * FROM departments;
23  SELECT * FROM employees;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

emp_id	emp_name	dept_id	salary
101	Rahul	1	54000.00
102	Priya	2	67000.00
103	Anil	2	55000.00
104	Sneha	3	71000.00
105	Kiran	2	48000.00
NULL	NULL	NULL	NULL

Result 35 | departments 36 | employees 37

Output

Action Output

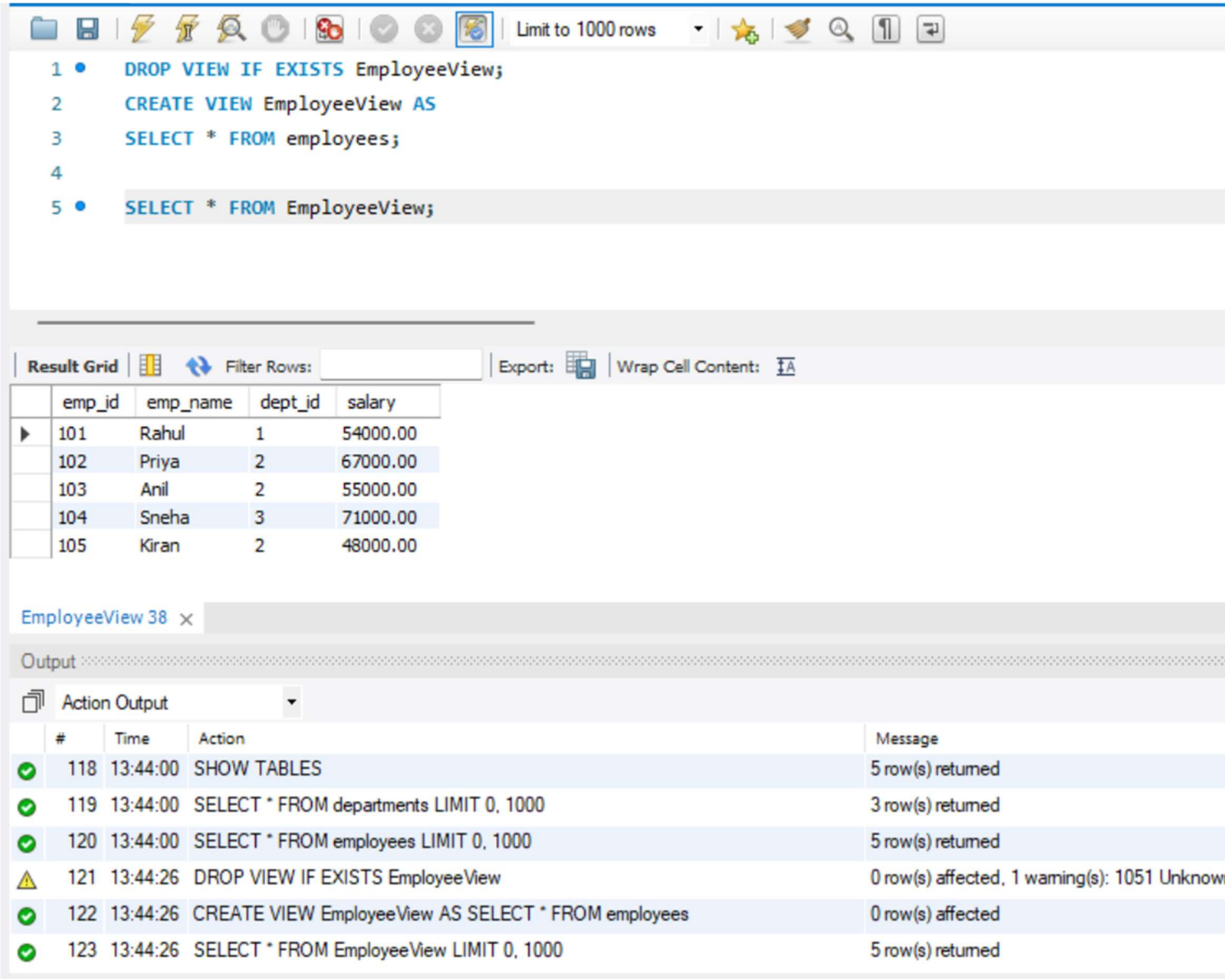
#	Time	Action	Message
115	13:44:00	CREATE TABLE IF NOT EXISTS employees (emp_id INT PRIMARY KEY, e...	0 row(s) affected, 1 warning(s): 1050 Table 'em
116	13:44:00	INSERT IGNORE INTO departments VALUES(1,'CSE'),(2,'ECE'),(3,'EEE')	0 row(s) affected, 3 warning(s): 1062 Duplicate
117	13:44:00	INSERT IGNORE INTO employees VALUES (101,'Rahul',1,50000),(102,'Priya',2,65...	0 row(s) affected, 5 warning(s): 1062 Duplicate
118	13:44:00	SHOW TABLES	5 row(s) returned
119	13:44:00	SELECT * FROM departments LIMIT 0, 1000	3 row(s) returned
120	13:44:00	SELECT * FROM employees LIMIT 0, 1000	5 row(s) returned

View 1 – Display All Employees

Question: Create a View named EmployeeView to display all columns from employees.

```
DROP VIEW IF EXISTS EmployeeView;

CREATE VIEW EmployeeView AS
SELECT *
FROM employees;
```

SELECT * FROM EmployeeView;



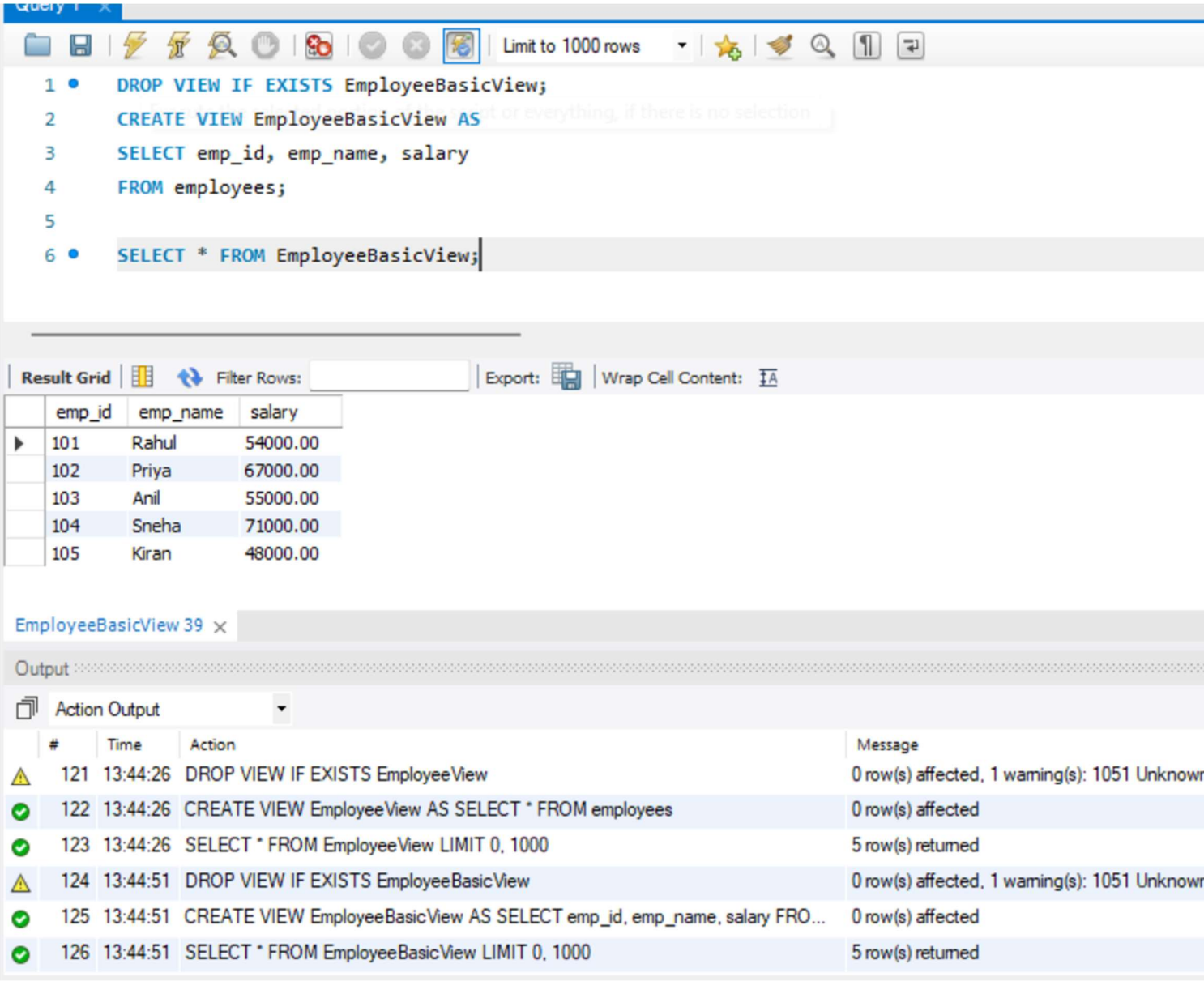
View 2 – Selected Columns

Question: Create a View showing employee ID, employee name and salary only.

DROP VIEW IF EXISTS EmployeeBasicView;

CREATE VIEW EmployeeBasicView AS
SELECT emp_id, emp_name, salary
FROM employees;

SELECT * FROM EmployeeBasicView;



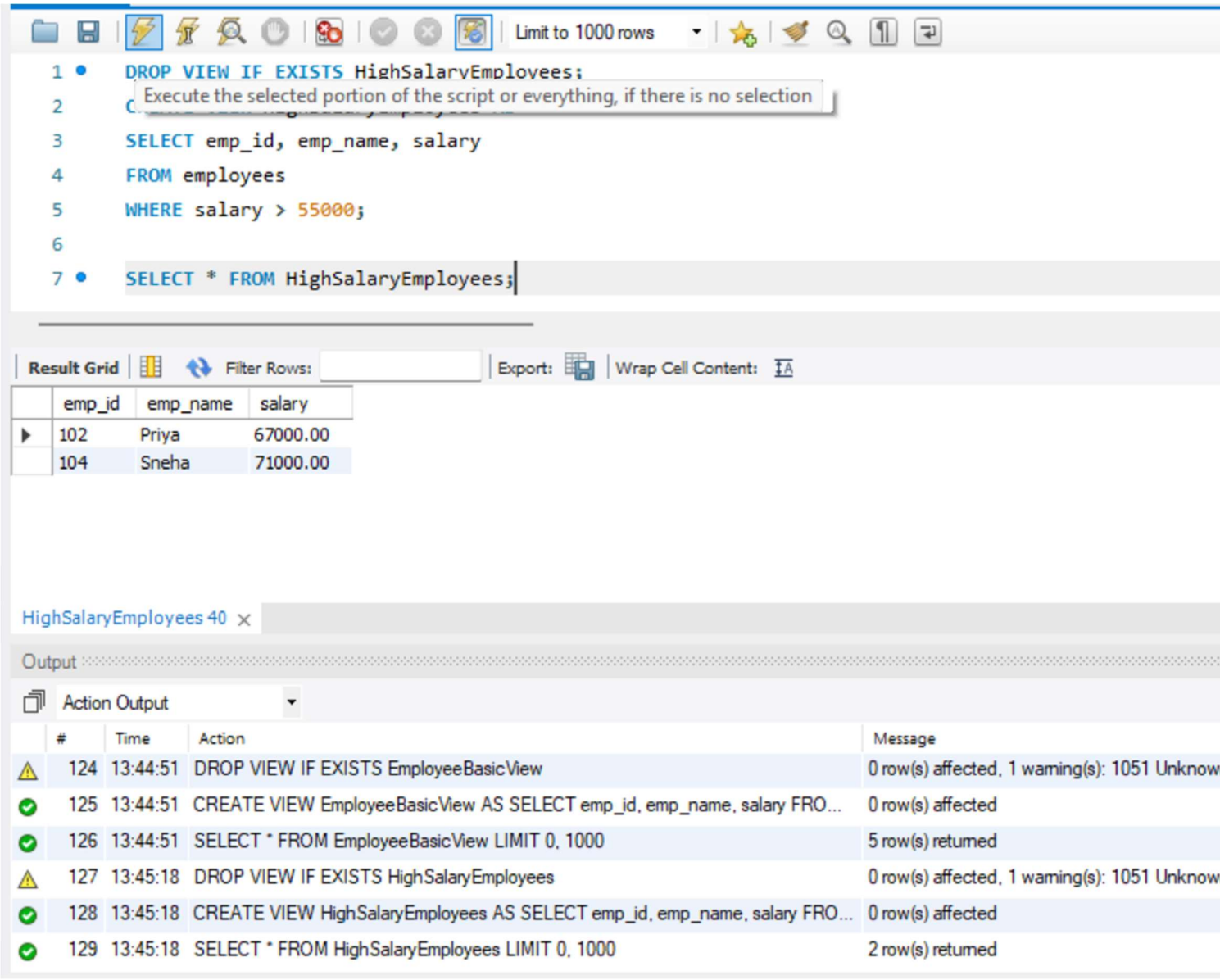
View 3 – High Salary Employees

Question: Create a View that displays employees earning more than 55000.

DROP VIEW IF EXISTS HighSalaryEmployees;

CREATE VIEW HighSalaryEmployees AS
SELECT emp_id, emp_name, salary
FROM employees
WHERE salary > 55000;

SELECT * FROM HighSalaryEmployees;



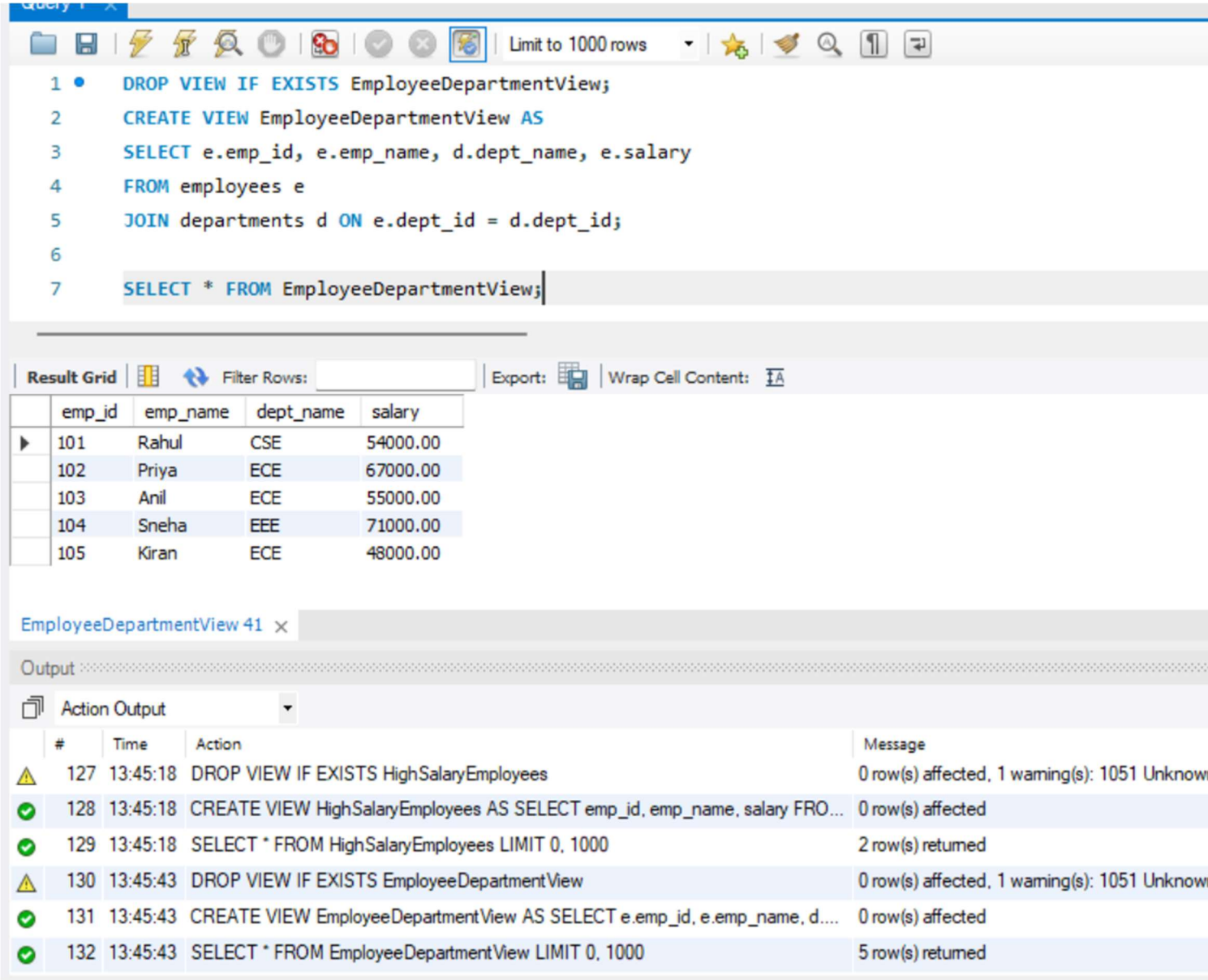
View 4 – Employee and Department using JOIN

Question: Create a View that displays employee name, department name and salary.

```
DROP VIEW IF EXISTS EmployeeDepartmentView;

CREATE VIEW EmployeeDepartmentView AS
SELECT
    e.emp_id,
    e.emp_name,
    d.dept_name,
    e.salary
FROM employees e
JOIN departments d
```

ON e.dept_id = d.dept_id;

SELECT * FROM EmployeeDepartmentView;



View 5 – Employees from a Particular Department

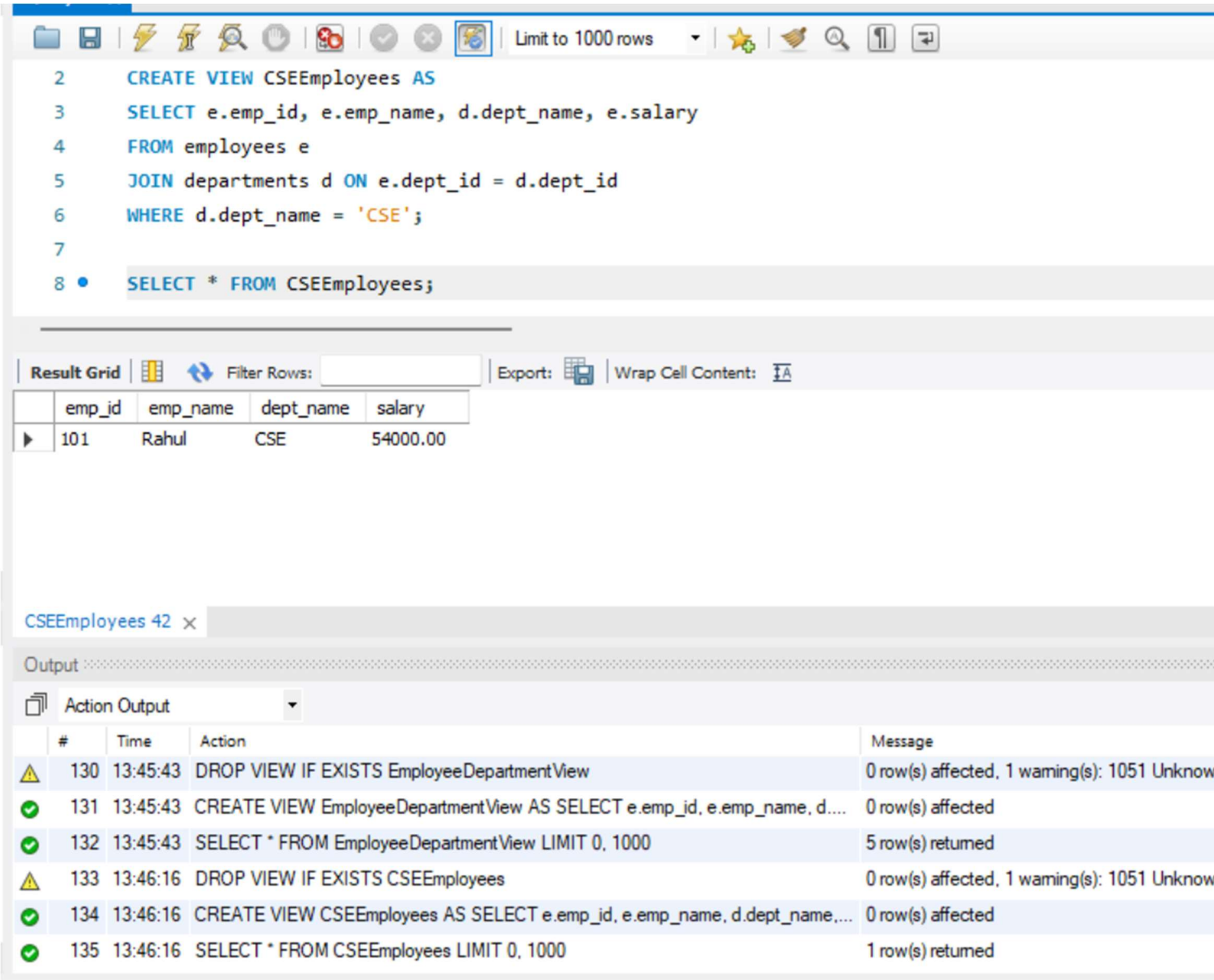
Question: Create a View that displays only CSE employees.

DROP VIEW IF EXISTS CSEEmployees;

CREATE VIEW CSEEmployees AS
SELECT
    e.emp_id,
    e.emp_name,
    d.dept_name,
    e.salary

FROM employees e
JOIN departments d
ON e.dept_id = d.dept_id
WHERE d.dept_name = 'CSE';

SELECT * FROM CSEEmployees;



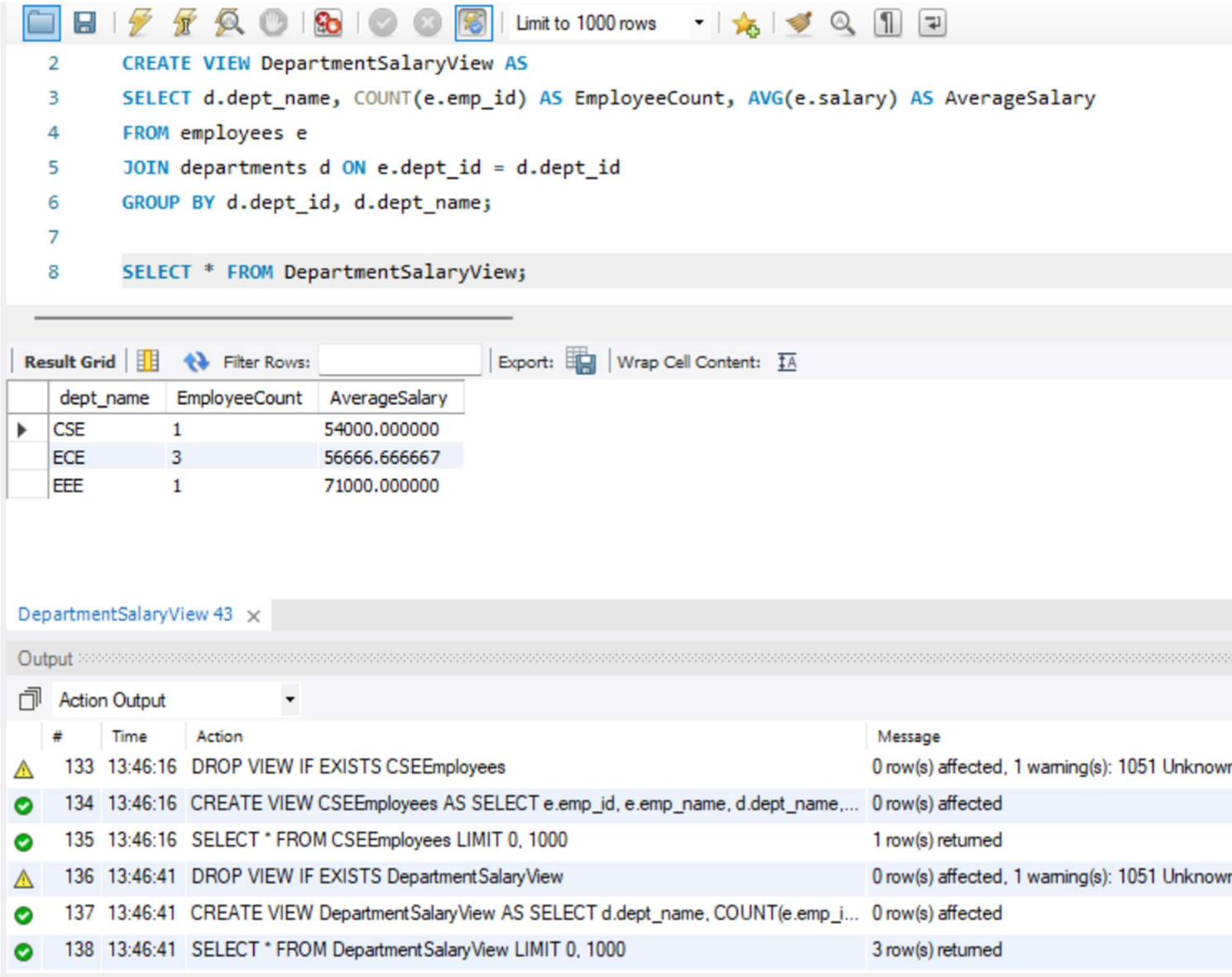
View 6 – Department-wise Average Salary

Question: Create a View using GROUP BY and AVG() to display department-wise average salary.

DROP VIEW IF EXISTS DepartmentSalaryView;

CREATE VIEW DepartmentSalaryView AS
SELECT
    d.dept_name,

```
    COUNT(e.emp_id) AS EmployeeCount,
    AVG(e.salary) AS AverageSalary
FROM employees e
JOIN departments d
ON e.dept_id = d.dept_id
GROUP BY d.dept_id, d.dept_name;

SELECT * FROM DepartmentSalaryView;
```

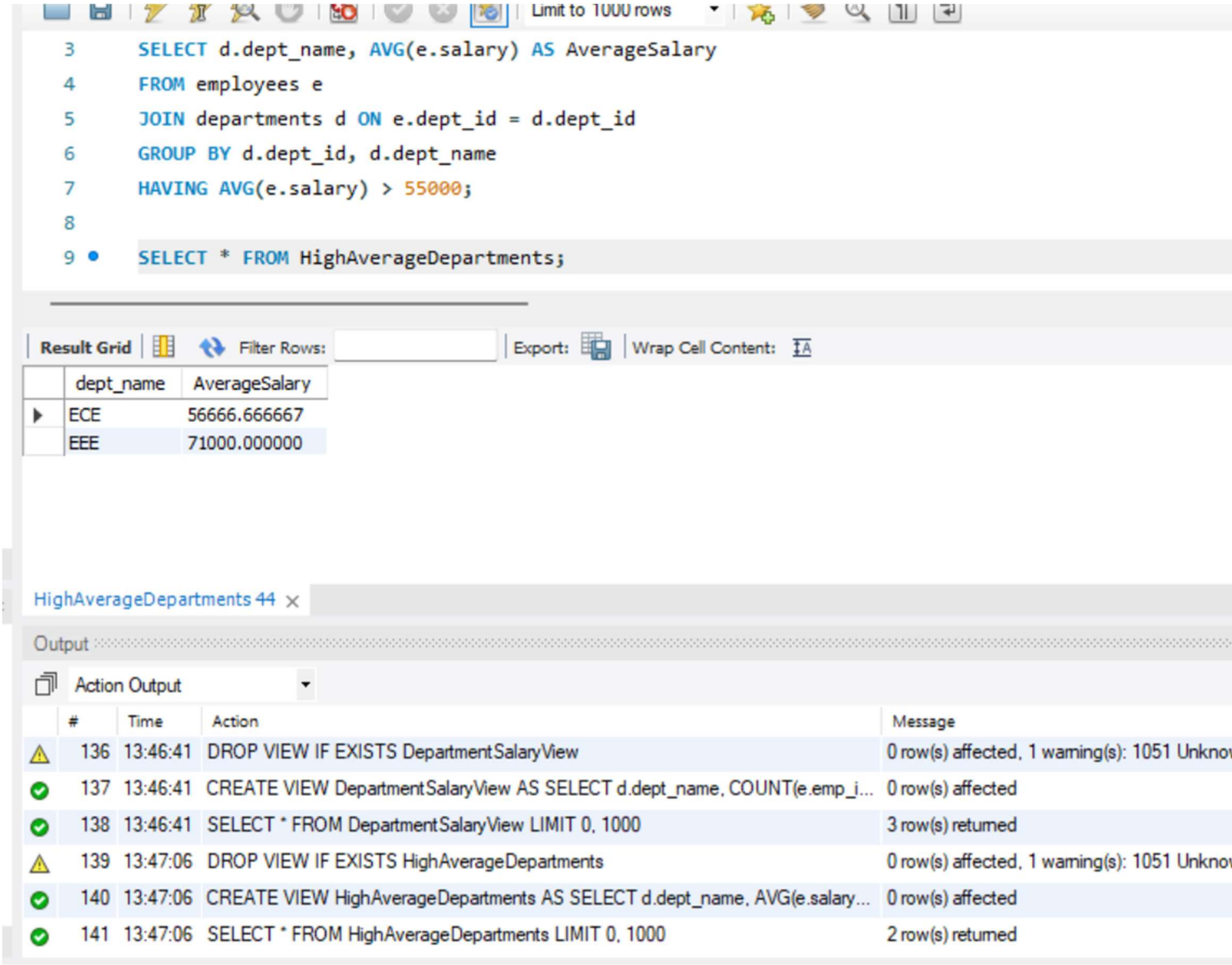


View 7 – Departments with Average Salary Above 55000

Question: Create a View that displays departments whose average salary is greater than 55000.

```
DROP VIEW IF EXISTS HighAverageDepartments;

CREATE VIEW HighAverageDepartments AS
```

```
SELECT
    d.dept_name,
    AVG(e.salary) AS AverageSalary
FROM employees e
JOIN departments d
ON e.dept_id = d.dept_id
GROUP BY d.dept_id, d.dept_name
HAVING AVG(e.salary) > 55000;

SELECT * FROM HighAverageDepartments;
```

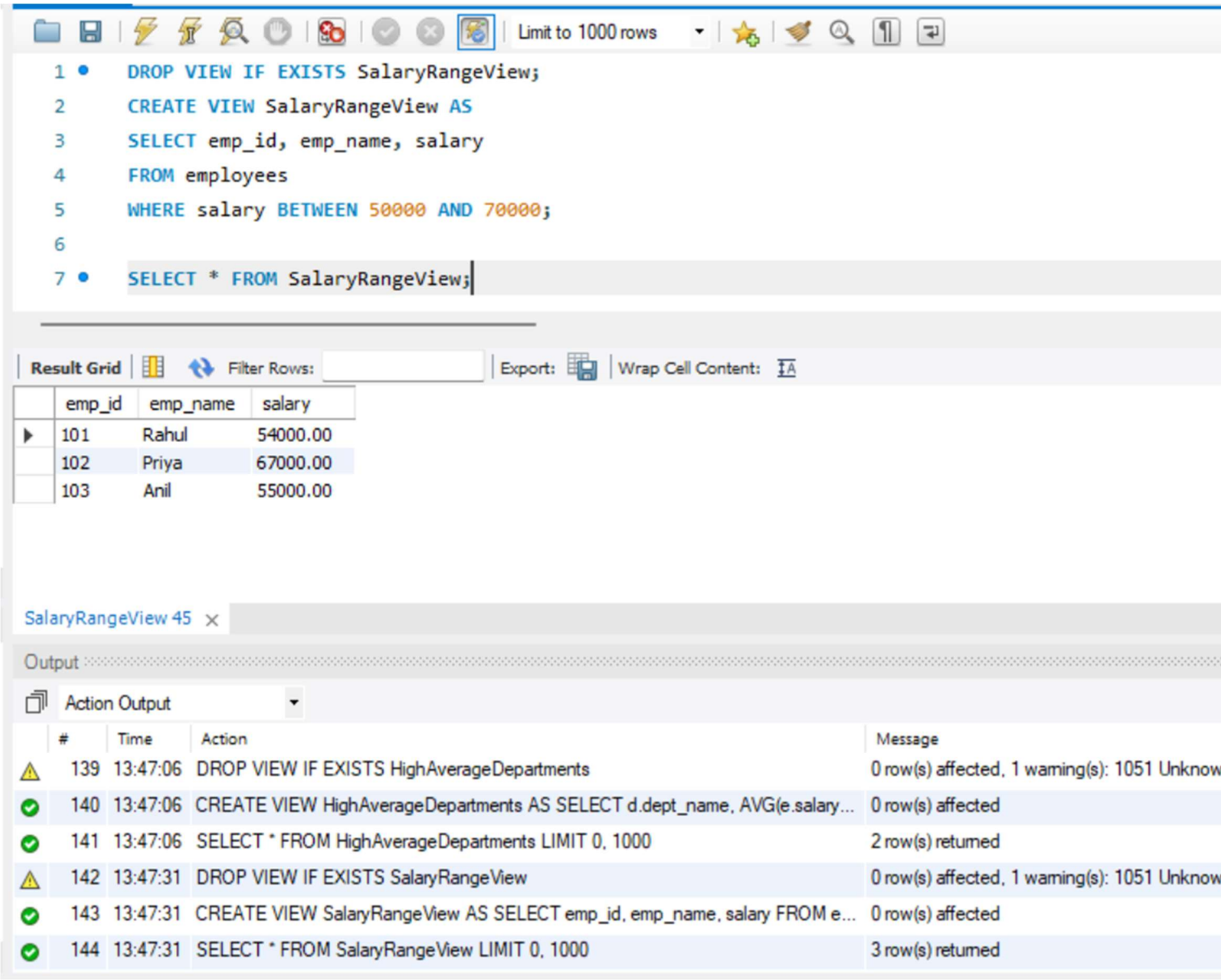


View 8 – Salary Range View

Question: Create a View that displays employees with salary between 50000 and 70000.

```
DROP VIEW IF EXISTS SalaryRangeView;
```

```
CREATE VIEW SalaryRangeView AS
SELECT emp_id, emp_name, salary
FROM employees
WHERE salary BETWEEN 50000 AND 70000;

SELECT * FROM SalaryRangeView;
```

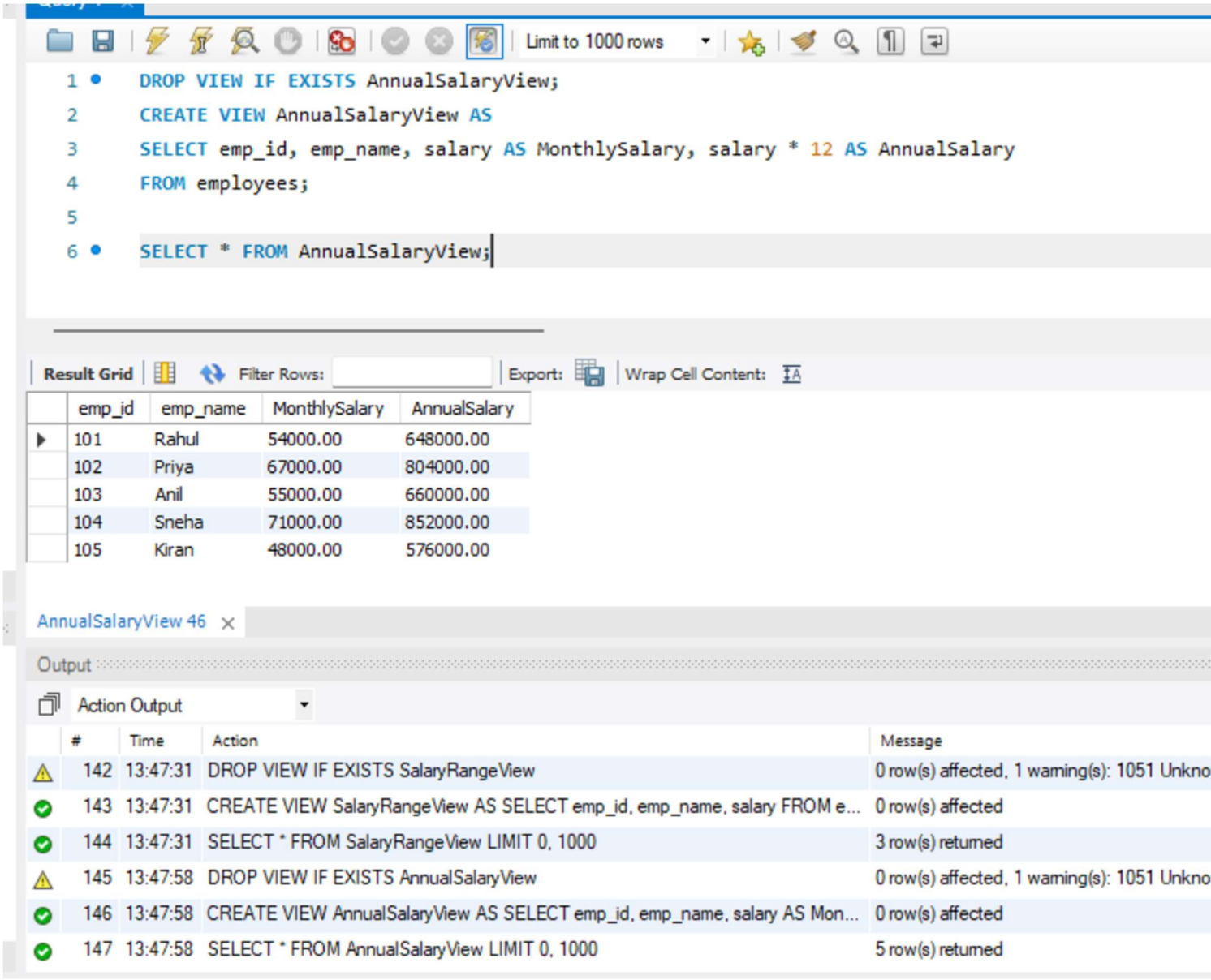


View 9 – View with Calculated Column

Question: Create a View that displays employee name, monthly salary and annual salary.

```
DROP VIEW IF EXISTS AnnualSalaryView;

CREATE VIEW AnnualSalaryView AS
SELECT
   emp_id,
```

```
    emp_name,
    salary AS MonthlySalary,
    salary * 12 AS AnnualSalary
FROM employees;

SELECT * FROM AnnualSalaryView;
```

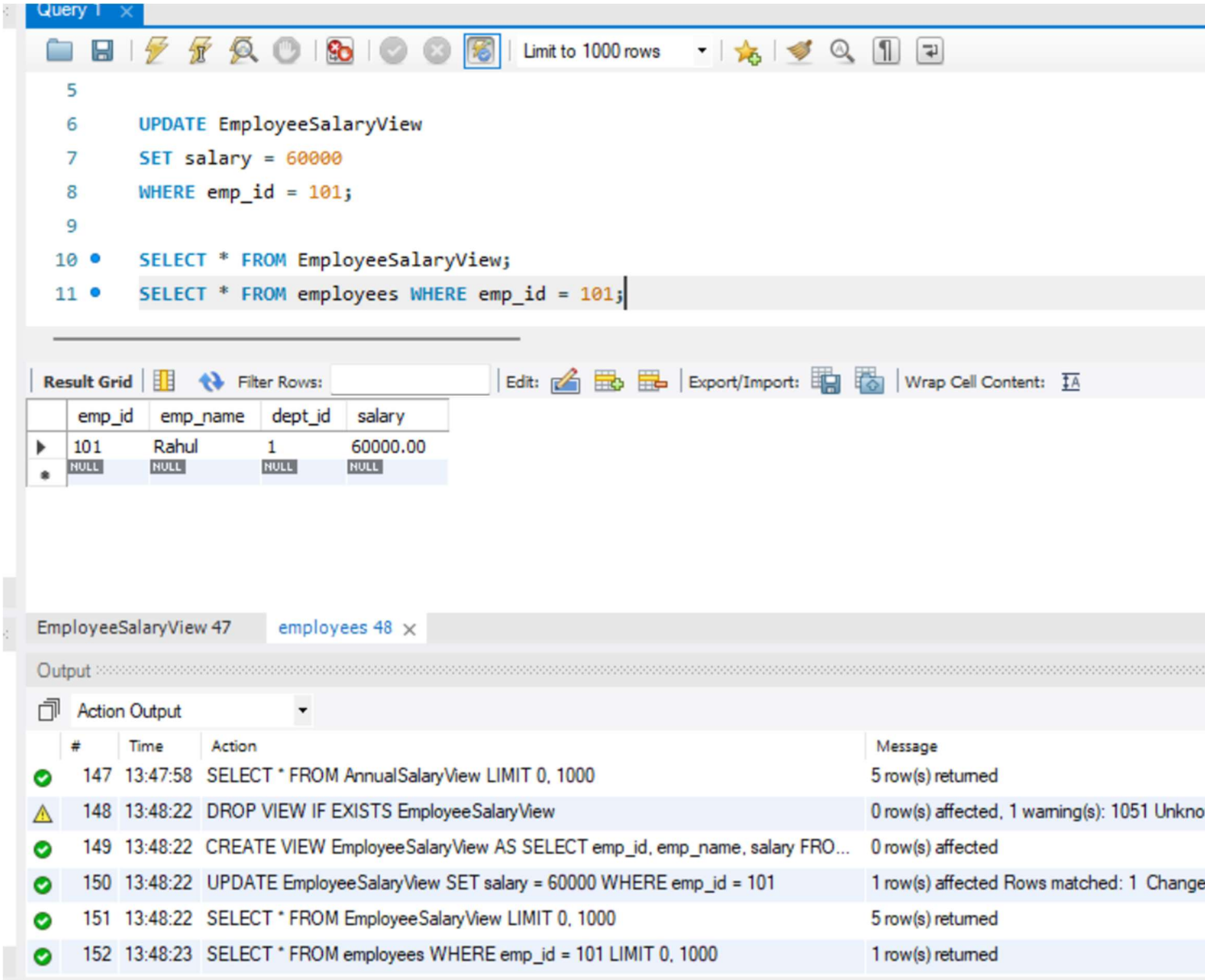


View 10 – Update Data Through a Simple View

Question: Create a simple single-table View and update an employee's salary through the View.

```
DROP VIEW IF EXISTS EmployeeSalaryView;

CREATE VIEW EmployeeSalaryView AS
SELECT emp_id, emp_name, salary
FROM employees;
```

```
UPDATE EmployeeSalaryView
SET salary = 60000
WHERE emp_id = 101;

SELECT * FROM EmployeeSalaryView;
SELECT * FROM employees WHERE emp_id = 101;
```

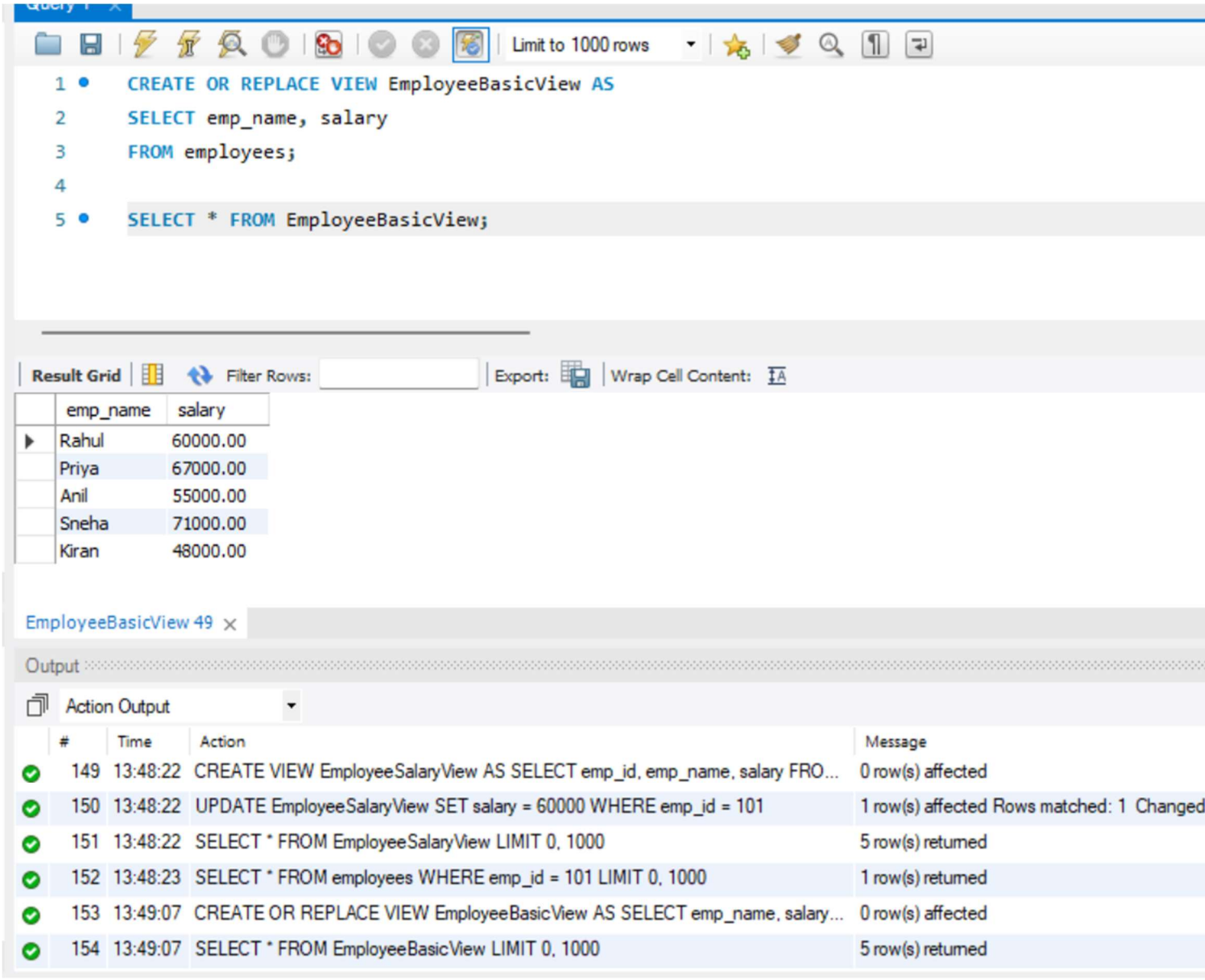


View 11 – CREATE OR REPLACE VIEW

Question: Modify an existing View so that it displays employee name and salary only.

```
CREATE OR REPLACE VIEW EmployeeBasicView AS
SELECT emp_name, salary
FROM employees;
```

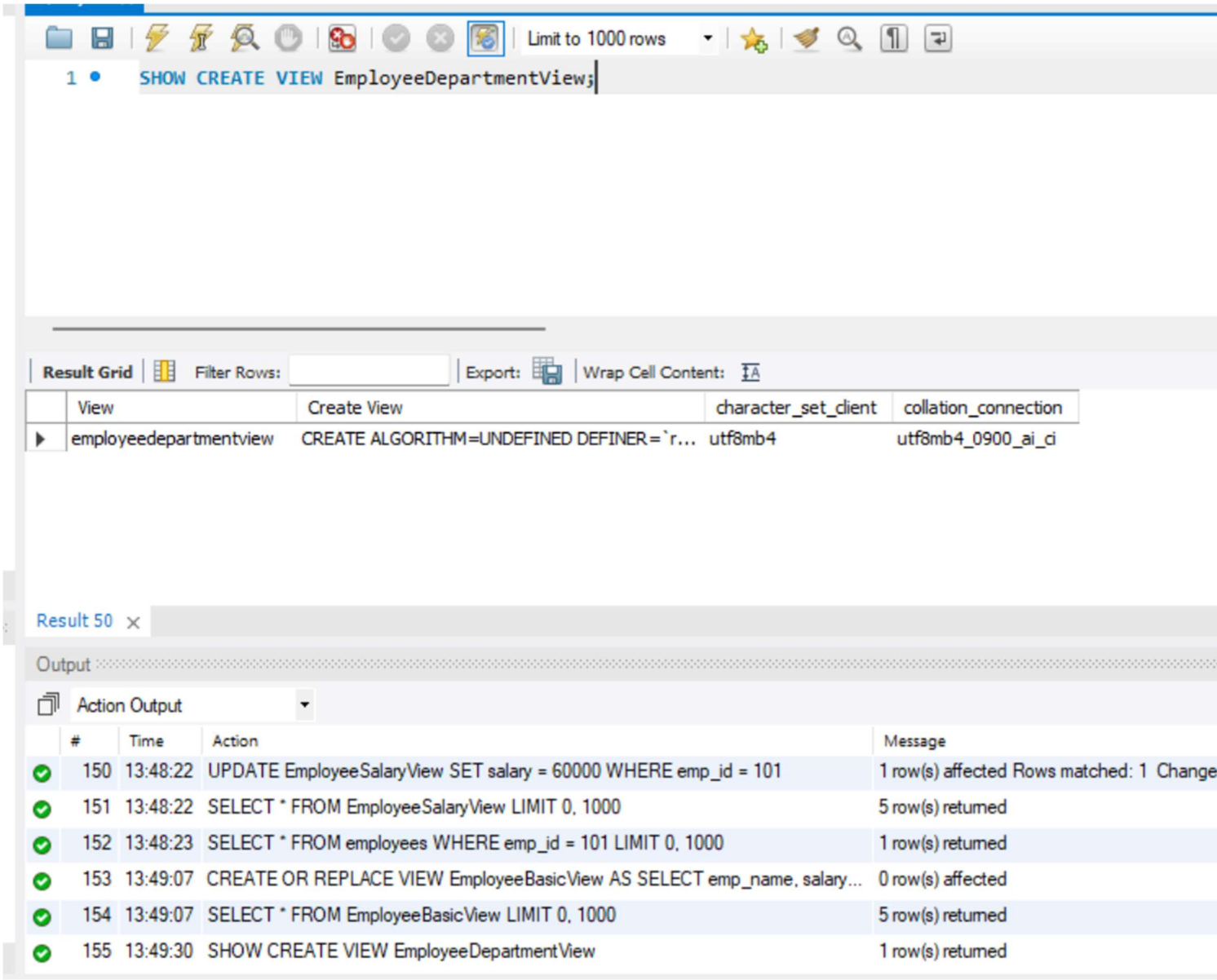
SELECT * FROM EmployeeBasicView;



View 12 – View Definition

Question: Display the SQL statement used to create a View.

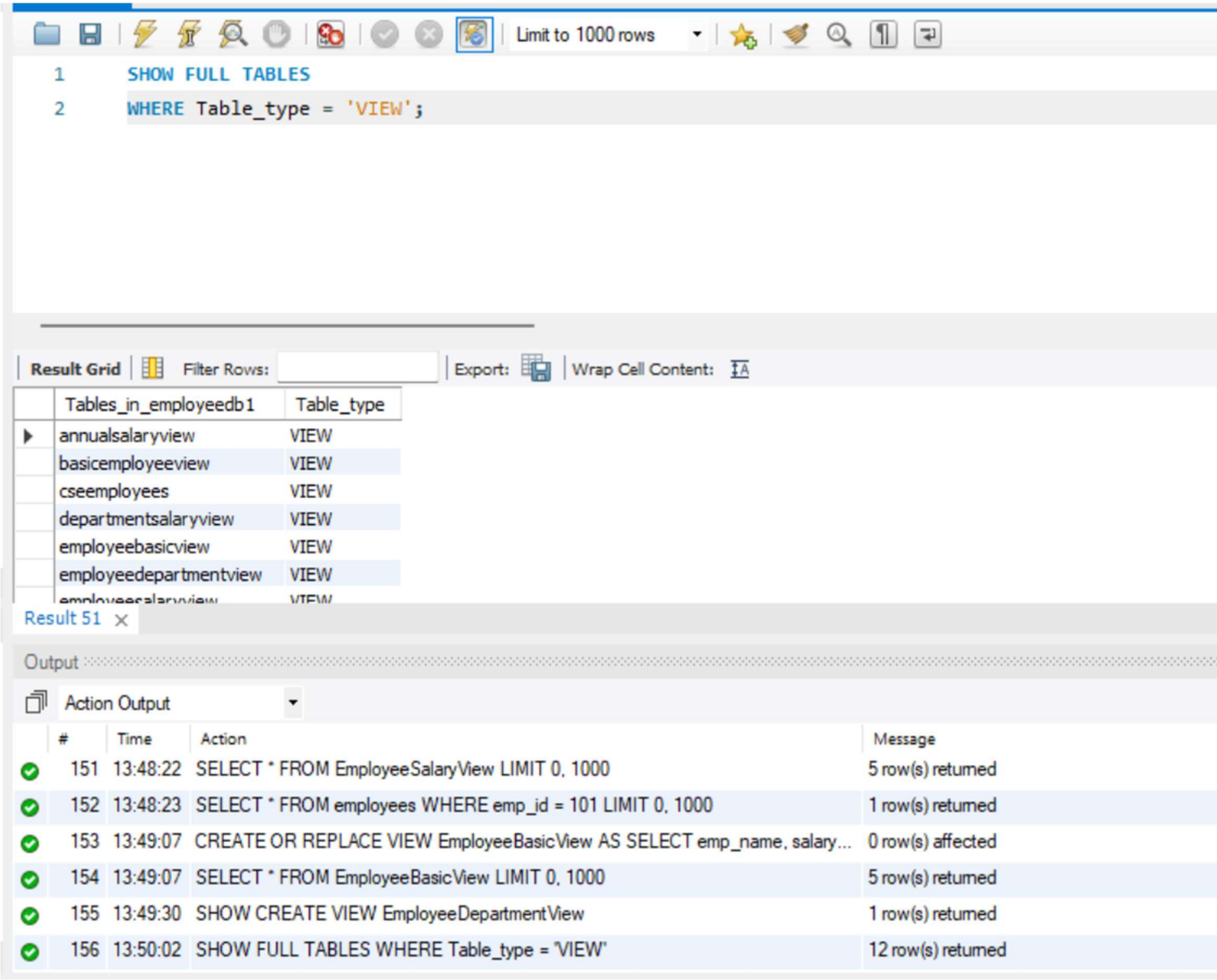
SHOW CREATE VIEW EmployeeDepartmentView;



View 13 – List All Views

Question: Display all Views available in the current database.

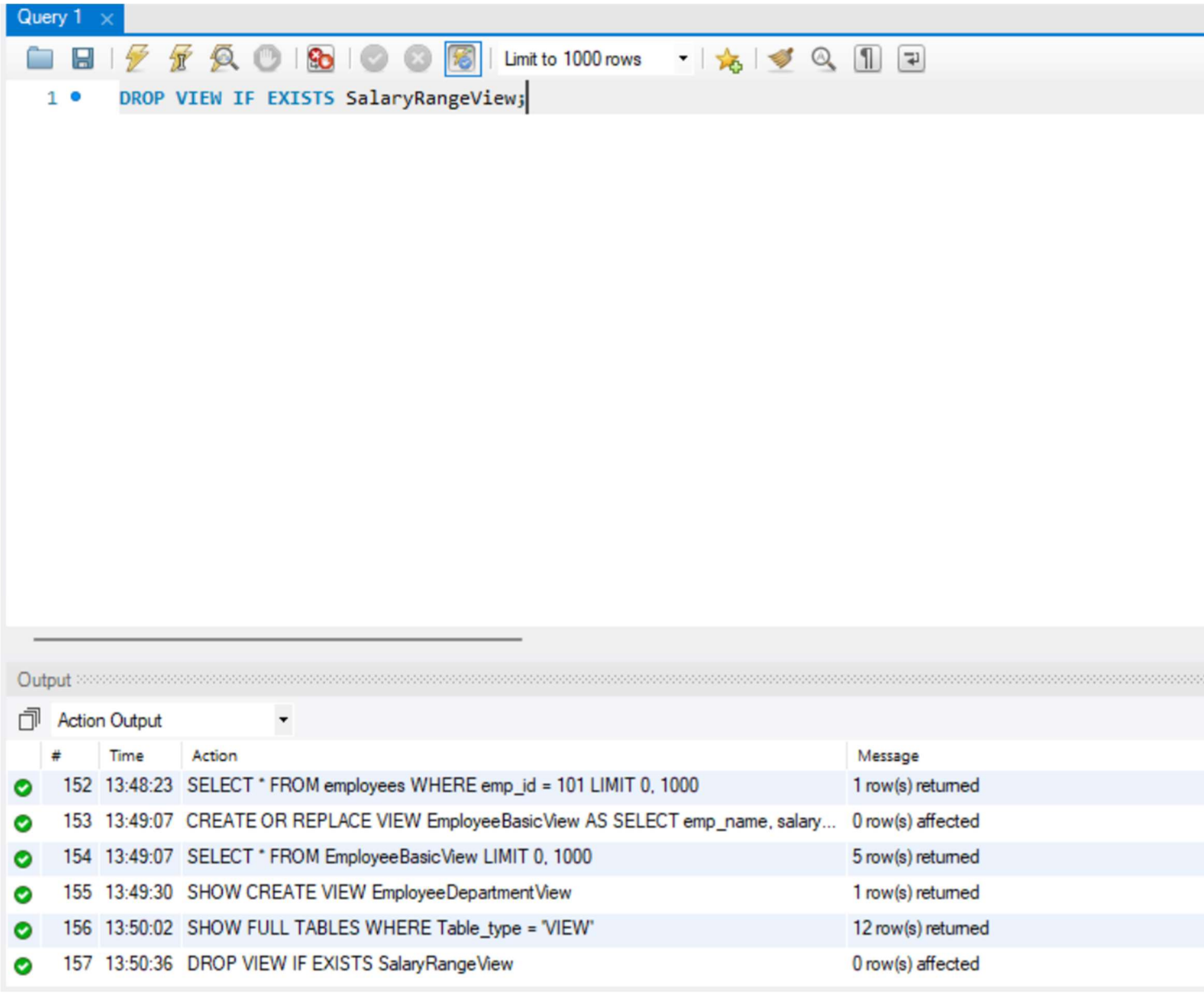
SHOW FULL TABLES
WHERE Table_type = 'VIEW';



View 14 – Drop a View

Question: Remove an existing View from the database.

DROP VIEW IF EXISTS SalaryRangeView;



4. Important Concept: Simple View vs Complex View

A simple View is generally based on one table and does not use grouping or aggregate functions. Some simple Views can be updated. A View containing JOINs, GROUP BY, aggregate functions such as AVG(), DISTINCT, or other non-updatable constructs may not be directly updatable.

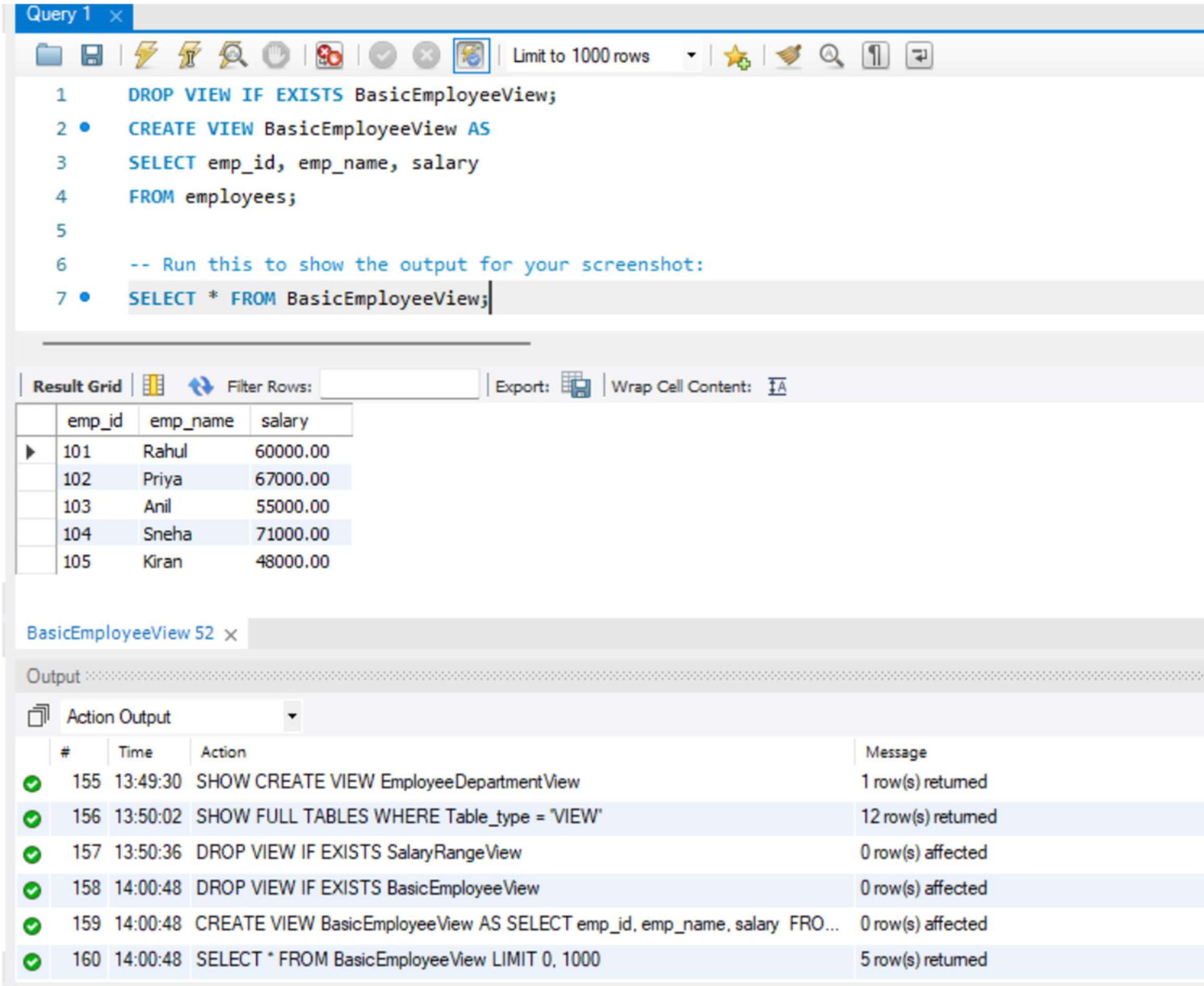
5. View vs Table

Feature	Table	View
Stores data	Yes	Normally no; stores query definition
Created with CREATE TABLE	Yes	No
Created with CREATE VIEW	No	Yes
Can use JOINs	Not applicable	Yes

Reflects current underlying data	Own stored data	Yes
Can always be updated	Yes	No; depends on the View

6. Student Practice Tasks

1. Create a View showing only emp_id, emp_name and salary.



2. Create a View for employees earning more than 60000.

```sql
DROP VIEW IF EXISTS HighEarnerView;
CREATE VIEW HighEarnerView AS
SELECT *
FROM employees
WHERE salary > 60000;

SELECT * FROM HighEarnerView;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_id	emp_name	dept_id	salary
102	Priya	2	67000.00
104	Sneha	3	71000.00

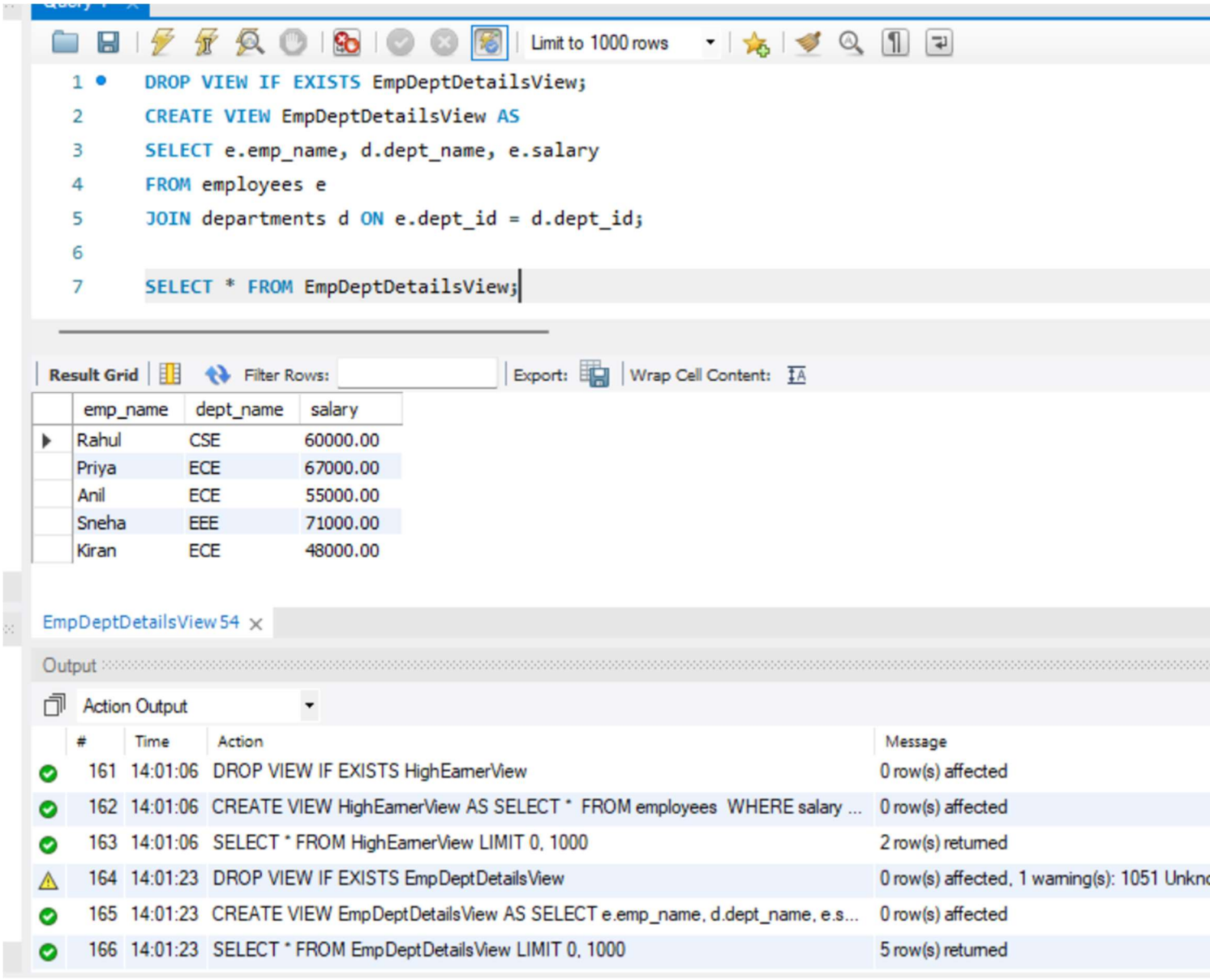
HighEarnerView 53

Output

Action Output

#	Time	Action	Message
158	14:00:48	DROP VIEW IF EXISTS BasicEmployeeView	0 row(s) affected
159	14:00:48	CREATE VIEW BasicEmployeeView AS SELECT emp_id, emp_name, salary FRO...	0 row(s) affected
160	14:00:48	SELECT * FROM BasicEmployeeView LIMIT 0, 1000	5 row(s) returned
161	14:01:06	DROP VIEW IF EXISTS HighEarnerView	0 row(s) affected
162	14:01:06	CREATE VIEW HighEarnerView AS SELECT * FROM employees WHERE salary ...	0 row(s) affected
163	14:01:06	SELECT * FROM HighEarnerView LIMIT 0, 1000	2 row(s) returned

3. Create a View showing employee name, department name and salary using JOIN.



```
DROP VIEW IF EXISTS EmpDeptDetailsView;
CREATE VIEW EmpDeptDetailsView AS
SELECT e.emp_name, d.dept_name, e.salary
FROM employees e
JOIN departments d ON e.dept_id = d.dept_id;

SELECT * FROM EmpDeptDetailsView;
```

Result Grid

emp_name	dept_name	salary
Rahul	CSE	60000.00
Priya	ECE	67000.00
Anil	ECE	55000.00
Sneha	EEE	71000.00
Kiran	ECE	48000.00

EmpDeptDetailsView 54

Output

Action Output

#	Time	Action	Message
161	14:01:06	DROP VIEW IF EXISTS HighEarnerView	0 row(s) affected
162	14:01:06	CREATE VIEW HighEarnerView AS SELECT * FROM employees WHERE salary ...	0 row(s) affected
163	14:01:06	SELECT * FROM HighEarnerView LIMIT 0, 1000	2 row(s) returned
164	14:01:23	DROP VIEW IF EXISTS EmpDeptDetailsView	0 row(s) affected, 1 warning(s): 1051 Unkno
165	14:01:23	CREATE VIEW EmpDeptDetailsView AS SELECT e.emp_name, d.dept_name, e.s...	0 row(s) affected
166	14:01:23	SELECT * FROM EmpDeptDetailsView LIMIT 0, 1000	5 row(s) returned

4. Create a View showing only ECE employees.

```
CREATE VIEW ECEEmployeesView AS
SELECT e.emp_id, e.emp_name, d.dept_name, e.salary
FROM employees e
JOIN departments d ON e.dept_id = d.dept_id
WHERE d.dept_name = 'ECE';

SELECT * FROM ECEEmployeesView;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_id	emp_name	dept_name	salary
102	Priya	ECE	67000.00
103	Anil	ECE	55000.00
105	Kiran	ECE	48000.00

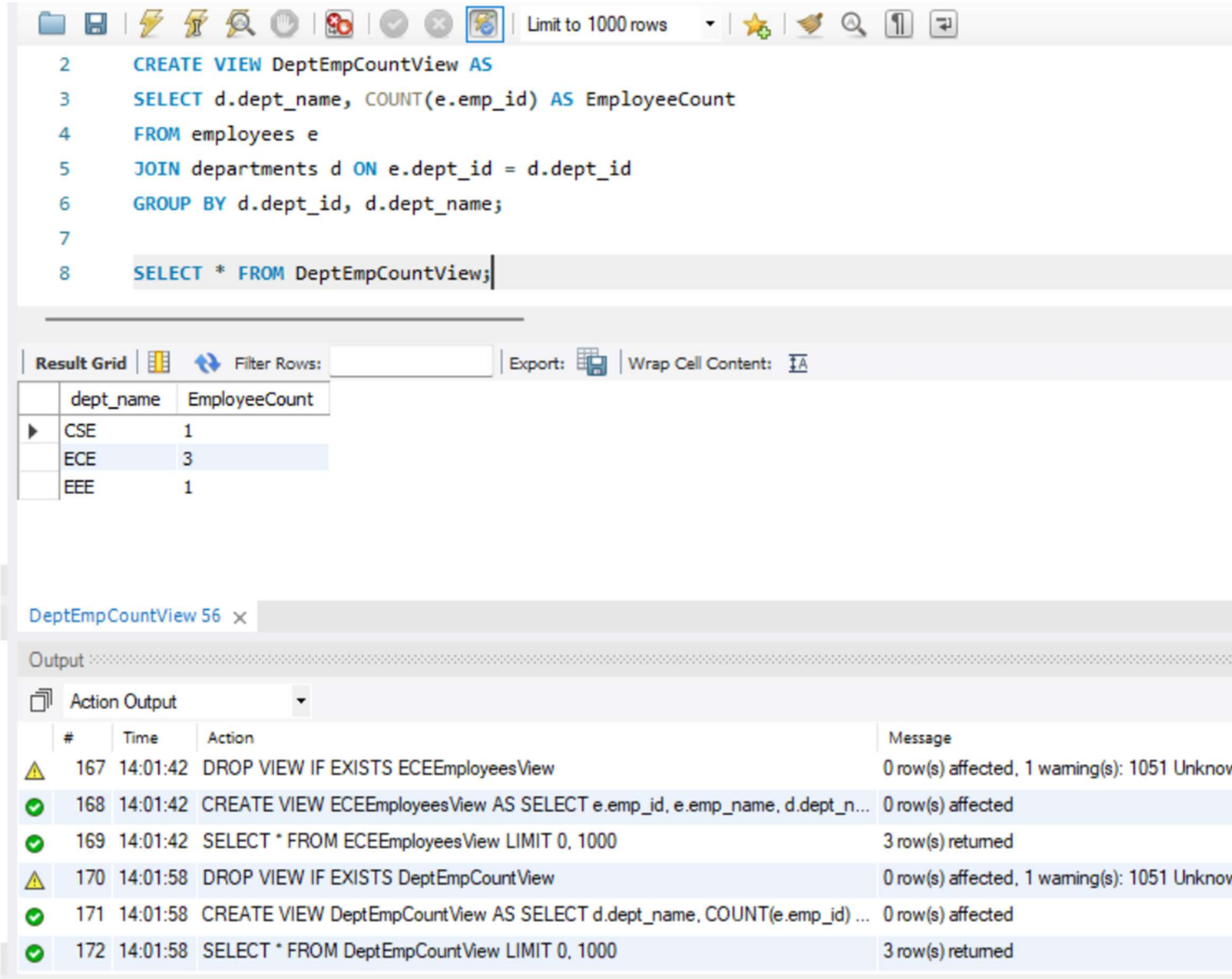
ECEEmployeesView 55

Output

Action Output

#	Time	Action	Message
164	14:01:23	DROP VIEW IF EXISTS EmpDeptDetailsView	0 row(s) affected, 1 warning(s): 1051 Unknown
165	14:01:23	CREATE VIEW EmpDeptDetailsView AS SELECT e.emp_name, d.dept_name, e.s...	0 row(s) affected
166	14:01:23	SELECT * FROM EmpDeptDetailsView LIMIT 0, 1000	5 row(s) returned
167	14:01:42	DROP VIEW IF EXISTS ECEEmployeesView	0 row(s) affected, 1 warning(s): 1051 Unknown
168	14:01:42	CREATE VIEW ECEEmployeesView AS SELECT e.emp_id, e.emp_name, d.dept_n...	0 row(s) affected
169	14:01:42	SELECT * FROM ECEEmployeesView LIMIT 0, 1000	3 row(s) returned

5. Create a View showing department-wise employee count.



```
2   CREATE VIEW DeptEmpCountView AS
3   SELECT d.dept_name, COUNT(e.emp_id) AS EmployeeCount
4   FROM employees e
5   JOIN departments d ON e.dept_id = d.dept_id
6   GROUP BY d.dept_id, d.dept_name;
7
8   SELECT * FROM DeptEmpCountView;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

dept_name	EmployeeCount
CSE	1
ECE	3
EEE	1

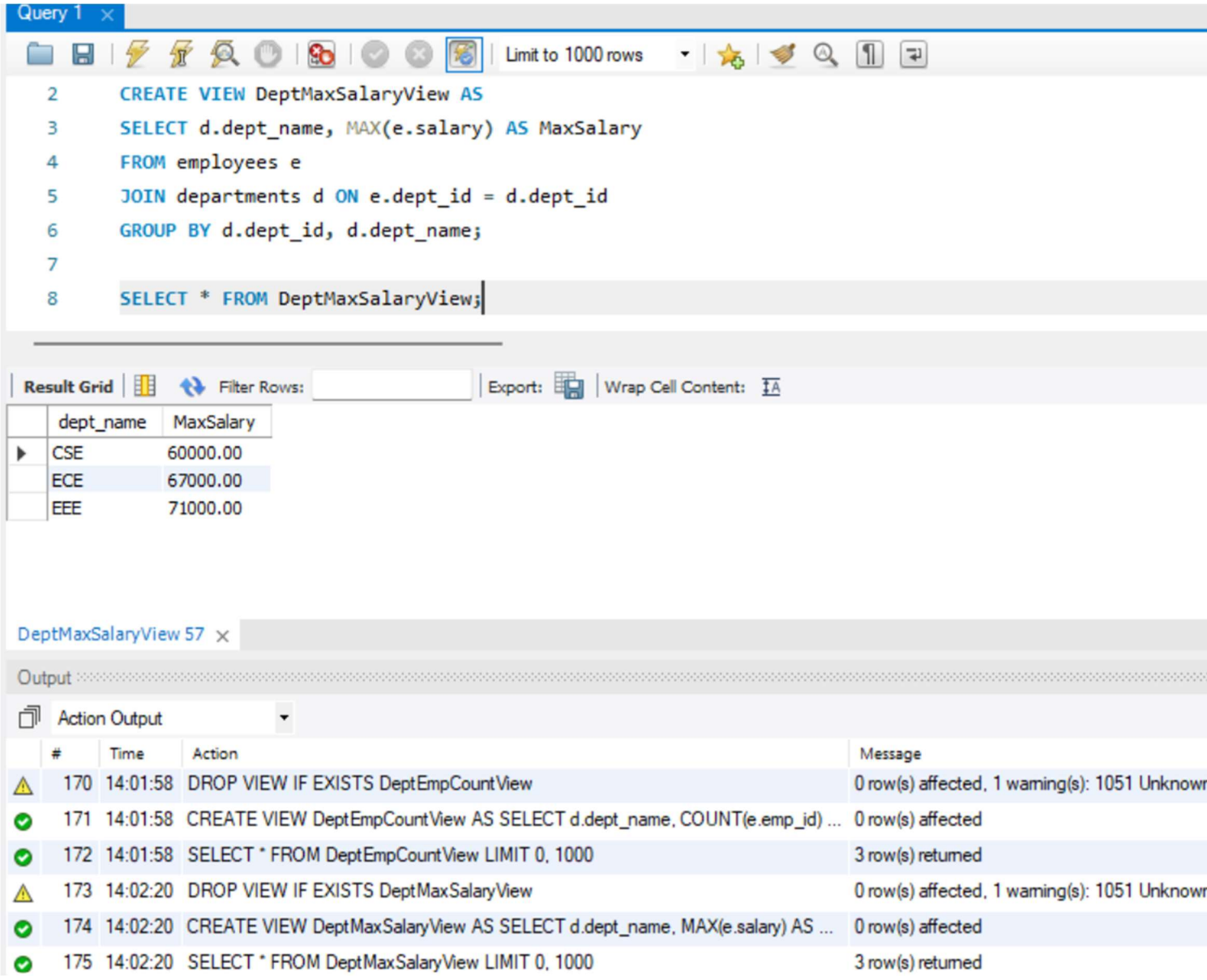
DeptEmpCountView 56

Output

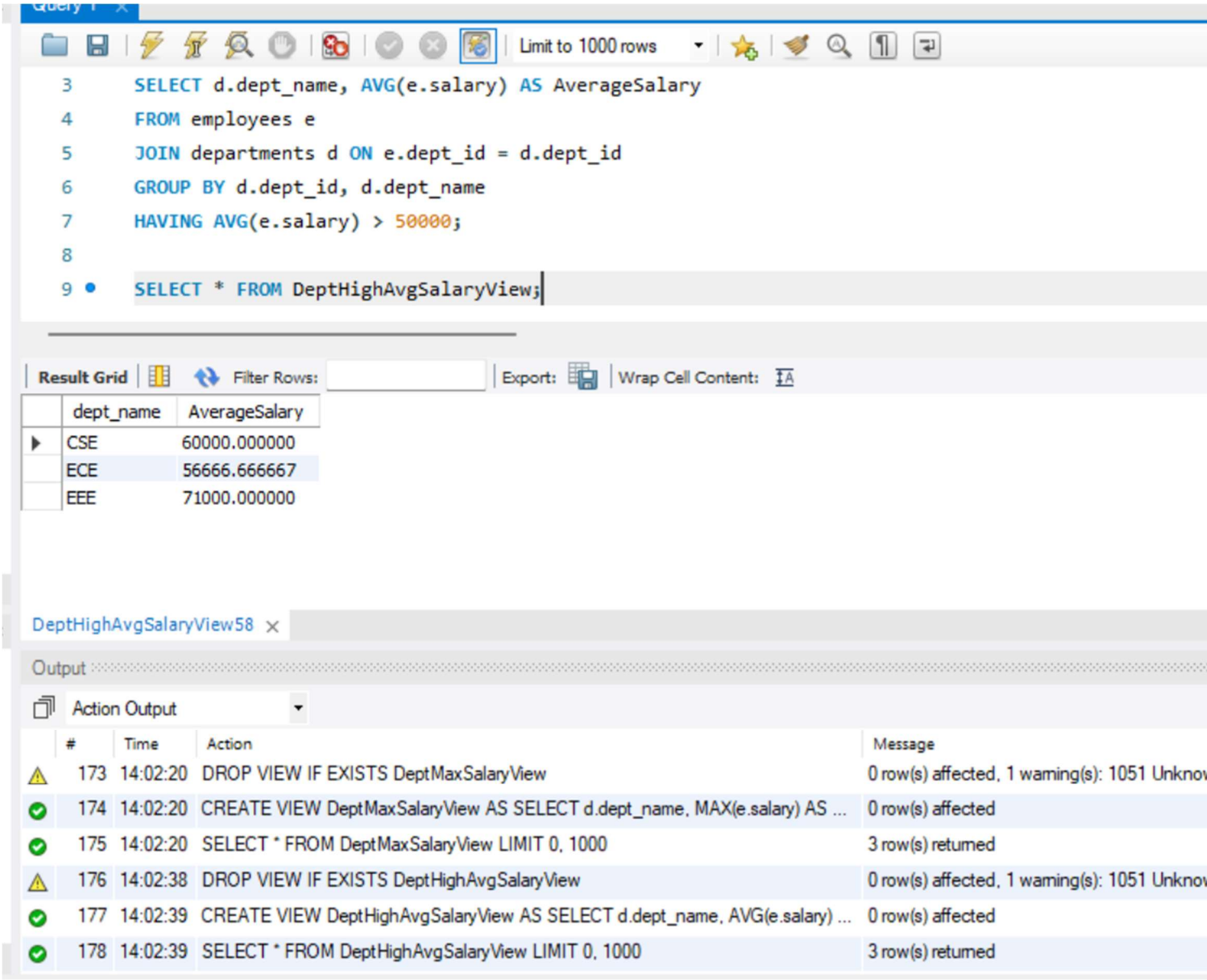
Action Output

#	Time	Action	Message
167	14:01:42	DROP VIEW IF EXISTS ECEEmployeesView	0 row(s) affected, 1 warning(s): 1051 Unknow
168	14:01:42	CREATE VIEW ECEEmployeesView AS SELECT e.emp_id, e.emp_name, d.dept_n...	0 row(s) affected
169	14:01:42	SELECT * FROM ECEEmployeesView LIMIT 0, 1000	3 row(s) returned
170	14:01:58	DROP VIEW IF EXISTS DeptEmpCountView	0 row(s) affected, 1 warning(s): 1051 Unknow
171	14:01:58	CREATE VIEW DeptEmpCountView AS SELECT d.dept_name, COUNT(e.emp_id) ...	0 row(s) affected
172	14:01:58	SELECT * FROM DeptEmpCountView LIMIT 0, 1000	3 row(s) returned

6. Create a View showing department-wise maximum salary.



7. Create a View showing departments whose average salary is greater than 50000.



```
SELECT d.dept_name, AVG(e.salary) AS AverageSalary
FROM employees e
JOIN departments d ON e.dept_id = d.dept_id
GROUP BY d.dept_id, d.dept_name
HAVING AVG(e.salary) > 50000;

SELECT * FROM DeptHighAvgSalaryView;
```

Result Grid

dept_name	AverageSalary
CSE	60000.000000
ECE	56666.666667
EEE	71000.000000

DeptHighAvgSalaryView58

Output

Action Output

#	Time	Action	Message
173	14:02:20	DROP VIEW IF EXISTS DeptMaxSalaryView	0 row(s) affected, 1 warning(s): 1051 Unknow
174	14:02:20	CREATE VIEW DeptMaxSalaryView AS SELECT d.dept_name, MAX(e.salary) AS ...	0 row(s) affected
175	14:02:20	SELECT * FROM DeptMaxSalaryView LIMIT 0, 1000	3 row(s) returned
176	14:02:38	DROP VIEW IF EXISTS DeptHighAvgSalaryView	0 row(s) affected, 1 warning(s): 1051 Unknow
177	14:02:39	CREATE VIEW DeptHighAvgSalaryView AS SELECT d.dept_name, AVG(e.salary) ...	0 row(s) affected
178	14:02:39	SELECT * FROM DeptHighAvgSalaryView LIMIT 0, 1000	3 row(s) returned

8. Create a View with employee name, salary and annual salary

.

```
1  DROP VIEW IF EXISTS AnnualSalaryCalcView;
2  CREATE VIEW AnnualSalaryCalcView AS
3  SELECT emp_name, salary AS MonthlySalary, (salary * 12) AS AnnualSalary
4  FROM employees;
5
6  SELECT * FROM AnnualSalaryCalcView;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_name	MonthlySalary	AnnualSalary
Rahul	60000.00	720000.00
Priya	67000.00	804000.00
Anil	55000.00	660000.00
Sneha	71000.00	852000.00
Kiran	48000.00	576000.00

AnnualSalaryCalcView 59

Output

Action Output

#	Time	Action	Message
176	14:02:38	DROP VIEW IF EXISTS DeptHighAvgSalaryView	0 row(s) affected, 1 warning(s): 1051 Unknow
177	14:02:39	CREATE VIEW DeptHighAvgSalaryView AS SELECT d.dept_name, AVG(e.salary) ...	0 row(s) affected
178	14:02:39	SELECT * FROM DeptHighAvgSalaryView LIMIT 0, 1000	3 row(s) returned
179	14:03:00	DROP VIEW IF EXISTS AnnualSalaryCalcView	0 row(s) affected, 1 warning(s): 1051 Unknow
180	14:03:00	CREATE VIEW AnnualSalaryCalcView AS SELECT emp_name, salary AS Monthly...	0 row(s) affected
181	14:03:00	SELECT * FROM AnnualSalaryCalcView LIMIT 0, 1000	5 row(s) returned

9. Update an employee's salary through a simple View and verify the base table.

```
-- This updates Rahul's salary using the view from Task 1
UPDATE BasicEmployeeView
SET salary = 52000
WHERE emp_id = 101;

-- This checks the base table to prove the update worked
SELECT * FROM employees WHERE emp_id = 101;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

emp_id	emp_name	dept_id	salary
101	Rahul	1	52000.00
NULL	NULL	NULL	NULL

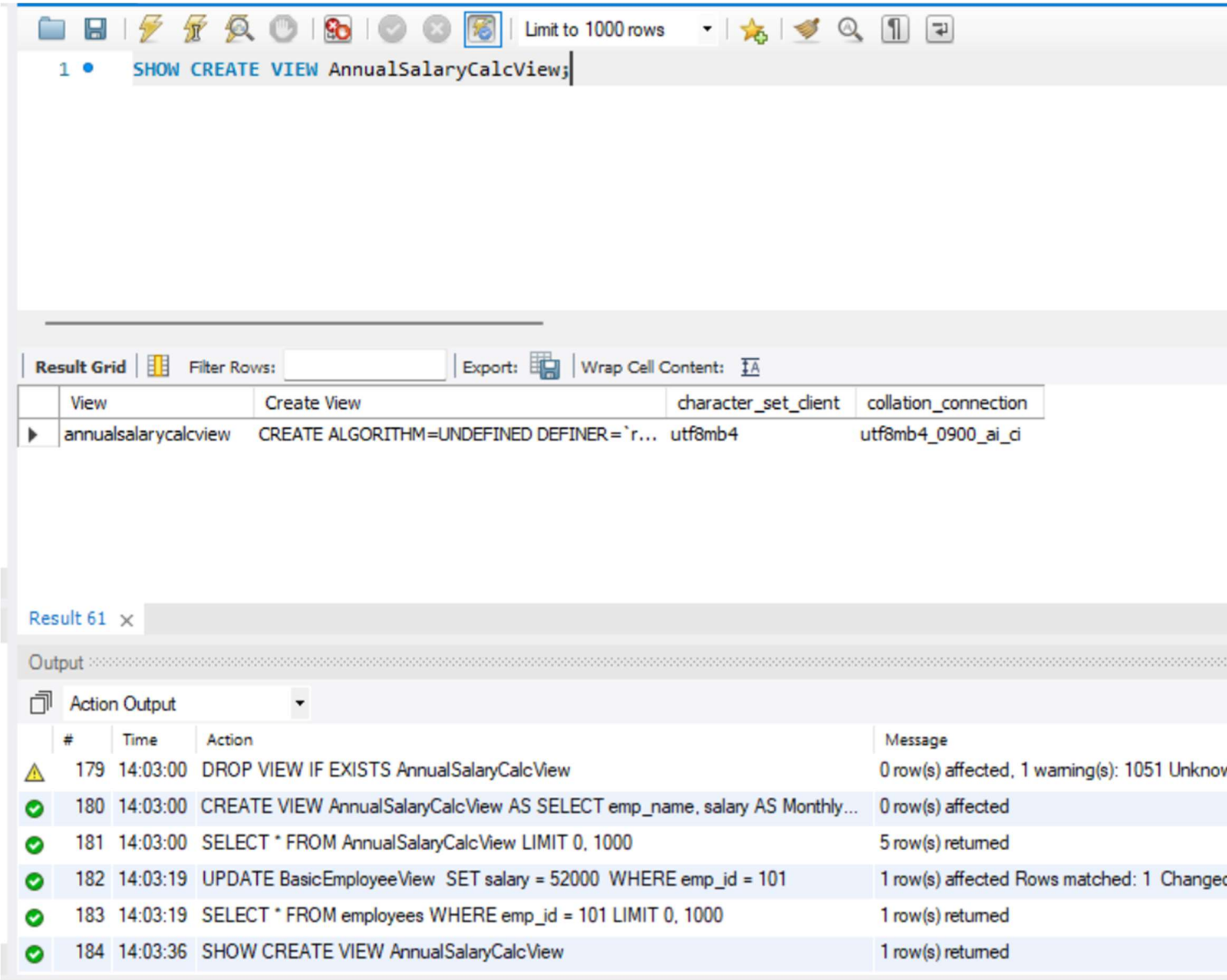
employees 60

Output

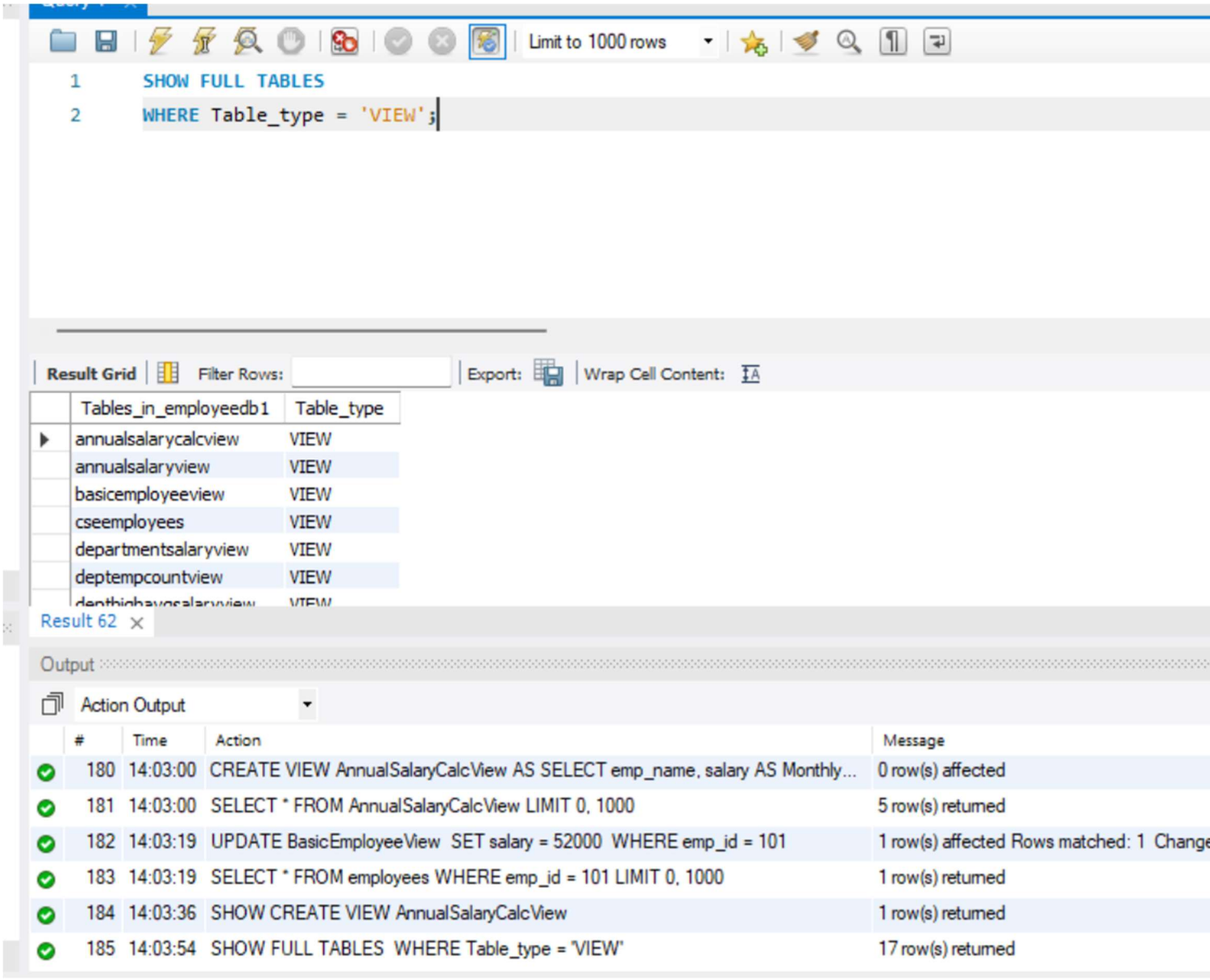
Action Output

#	Time	Action	Message
178	14:02:39	SELECT * FROM DeptHighAvgSalaryView LIMIT 0, 1000	3 row(s) returned
179	14:03:00	DROP VIEW IF EXISTS AnnualSalaryCalcView	0 row(s) affected, 1 warning(s): 1051 Unknow
180	14:03:00	CREATE VIEW AnnualSalaryCalcView AS SELECT emp_name, salary AS Monthly...	0 row(s) affected
181	14:03:00	SELECT * FROM AnnualSalaryCalcView LIMIT 0, 1000	5 row(s) returned
182	14:03:19	UPDATE BasicEmployeeView SET salary = 52000 WHERE emp_id = 101	1 row(s) affected Rows matched: 1 Changed
183	14:03:19	SELECT * FROM employees WHERE emp_id = 101 LIMIT 0, 1000	1 row(s) returned

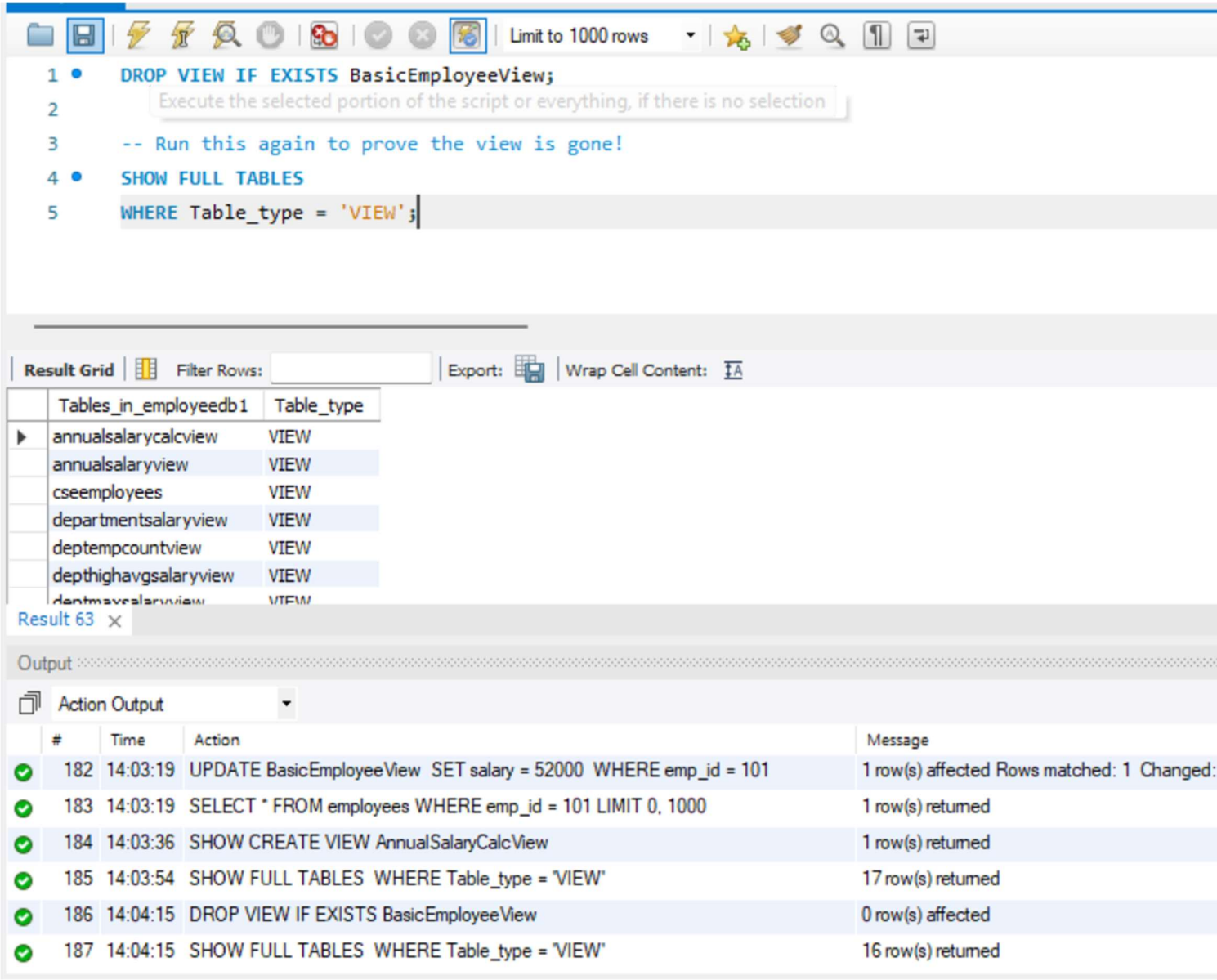
10. Use SHOW CREATE VIEW to display the definition of a View.



11. List all Views in the database.



12. Drop a View and verify that it no longer appears in the list.



7. Viva / Discussion Questions

1. What is a View?

A View is a virtual table created from one or more existing tables. It does not store data itself; it stores only the SQL query.

2. Why do we use Views?

Views are used to simplify complex queries, automatically calculate columns, filter specific data, and enhance database security.

3. Does a View normally store a separate copy of the data?

Normally no; a View stores the query definition and retrieves the latest data from the underlying tables

whenever you query it.

4. What is the difference between a View and a table?
A table physically stores its own data and is created with the CREATE TABLE command. A View normally does not store data, is created with the CREATE VIEW command, and can use JOINs to combine tables.

5. Can a View be created using multiple tables?
Yes, a View can be created from one or more existing tables, often by using JOIN statements.

6. Can every View be updated?
No, it depends on the View. Some simple Views can be updated, but a View containing JOINs, GROUP BY, aggregate functions (like AVG()), DISTINCT, or other non-updatable constructs may not be directly updatable.

7. Why are Views useful for security?
They allow you to restrict access by creating Views that only show specific columns or rows instead of giving a user full access to the entire base table.

8. What is the difference between CREATE VIEW and CREATE OR REPLACE VIEW?
CREATE VIEW is used to make a brand-new View, while CREATE OR REPLACE VIEW will modify an existing View without dropping it, or create it if it does not already exist.

9. How do you display the definition of a View in MySQL?
use the command SHOW CREATE VIEW view_name;.

10. How do you remove a View?
remove a View using the command DROP VIEW IF EXISTS view_name;.

8. Common Errors and Solutions

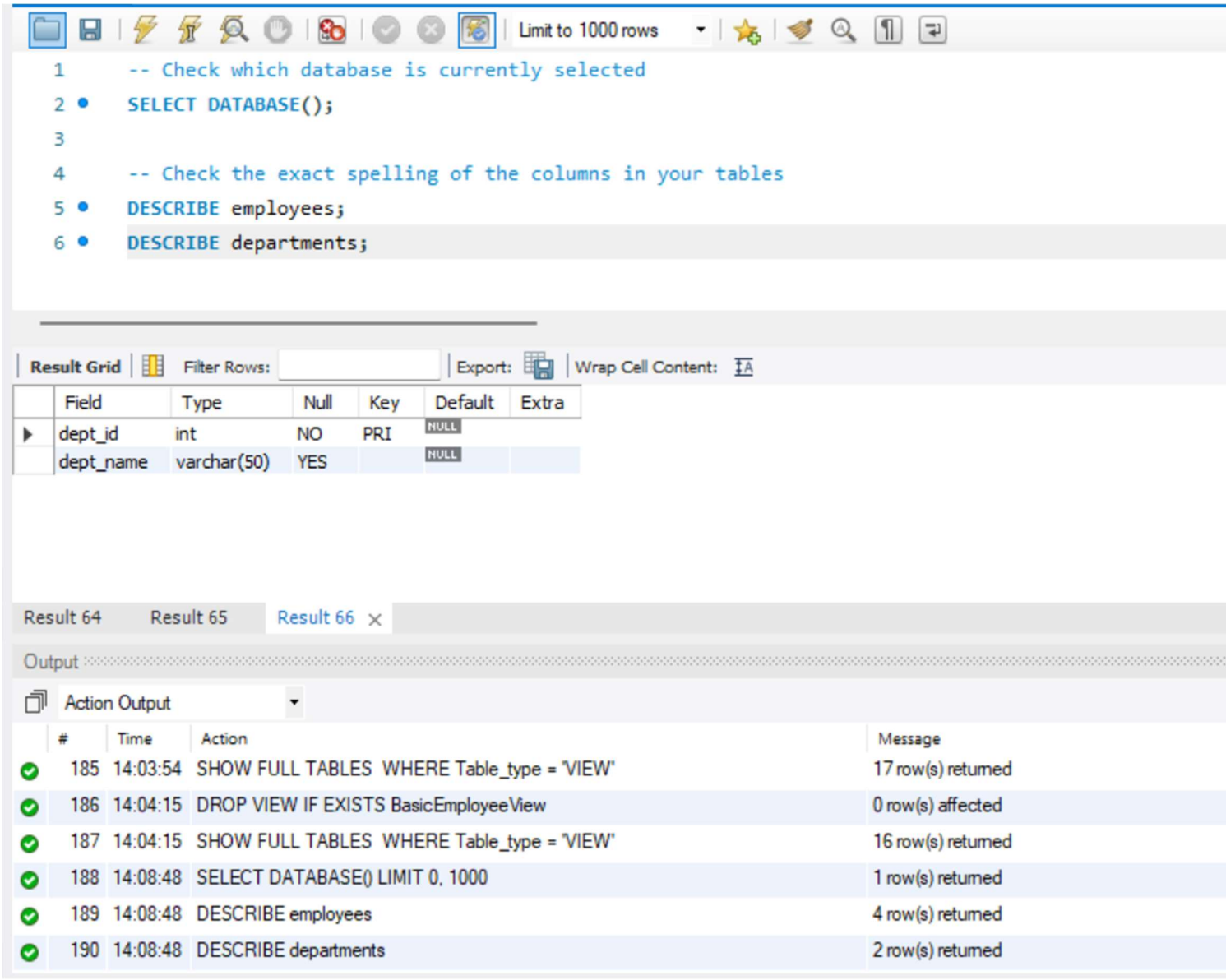
Table does not exist: Run USE employeedb1; and check with SHOW TABLES;.

View already exists: Use DROP VIEW IF EXISTS view_name; or CREATE OR REPLACE VIEW.

Unknown column: Check the column names using DESCRIBE employees; or DESCRIBE departments;.

Cannot update View: The View may contain JOIN, GROUP BY, aggregate functions or another non-updatable construct.

Wrong database: Check the selected schema and run SELECT DATABASE();.



9. Quick Reference

CREATE VIEW view_name AS
SELECT ...;

SELECT * FROM view_name;

CREATE OR REPLACE VIEW view_name AS
SELECT ...;

SHOW CREATE VIEW view_name;

SHOW FULL TABLES
WHERE Table_type='VIEW';

DROP VIEW IF EXISTS view_name;